Engineering Practice(EP)

UNIT- II

Carpentry shop

Introduction of carpentry/Joinery, wood working: types of woods, advantage of wood, seasoning of wood, carpentry tolls: measuring tools, marking tolls, cutting tools: saws, chisels, planning tools, drilling tools, striking tools, drilling tools, wood working joints, wood working lathe

Foundry Shop

Introduction, foundry hand tools, measuring boxes, ladle, moulding, machines, furnaces.

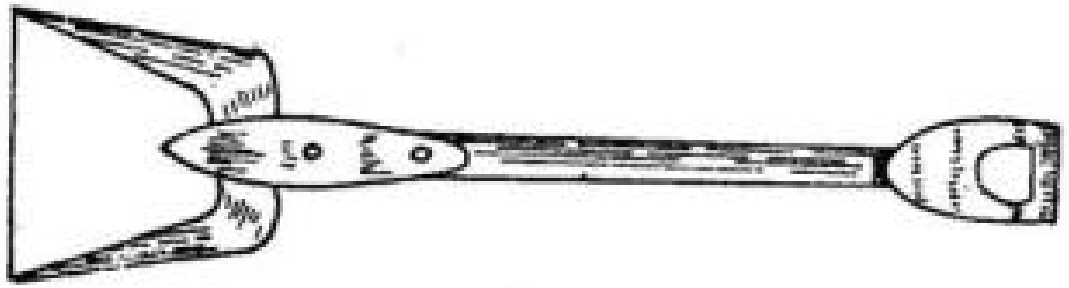
Foundry

A foundry is a **workshop in which casting of metal is done**.

Metals are cast into shapes by melting them into a liquid, pouring the metal into a mold, and removing the mold material after the metal has solidified as it cools. In this process, parts of desired shapes and sizes can be formed.

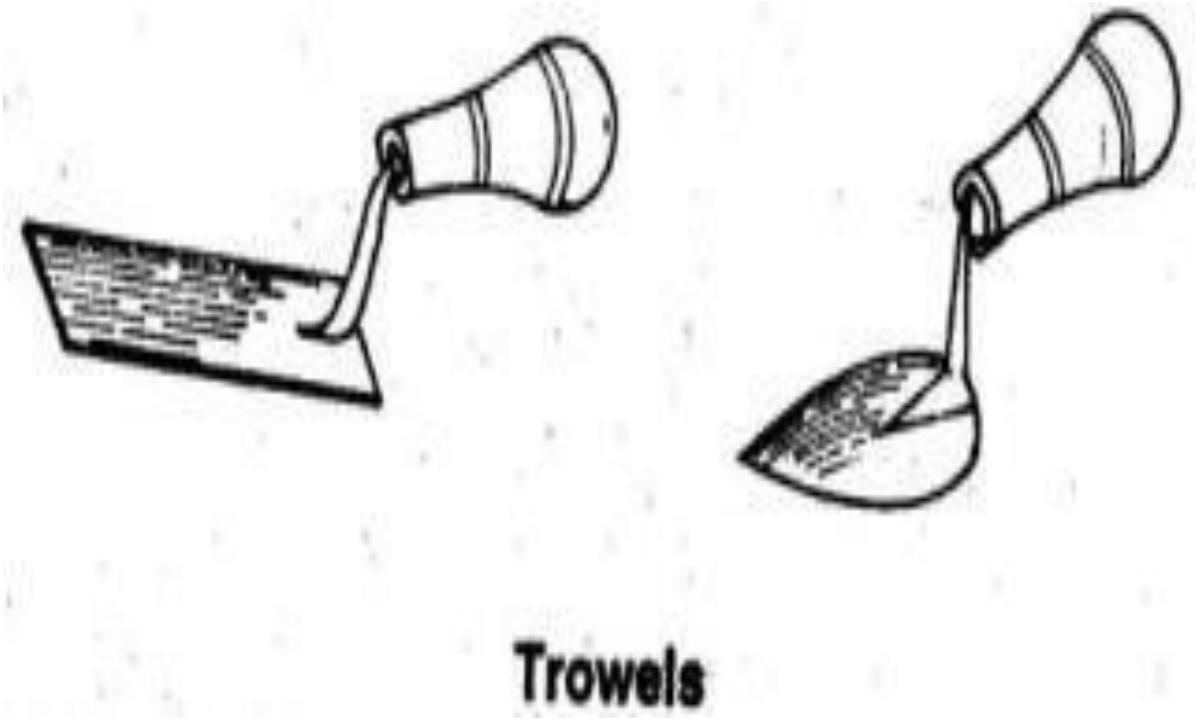
Foundry Hand Tools

Showel: It consists of iron pan with a wooden handle. It can be used for mixing and conditioning the sand.



Showel

Trowels: The trowels are basically employed for smoothing or slicking the surfaces of molds. They may also be used to cut in-gates and repair the mold surfaces. Common shapes of trowels are shown as under. They are made of iron with a wooden handle.



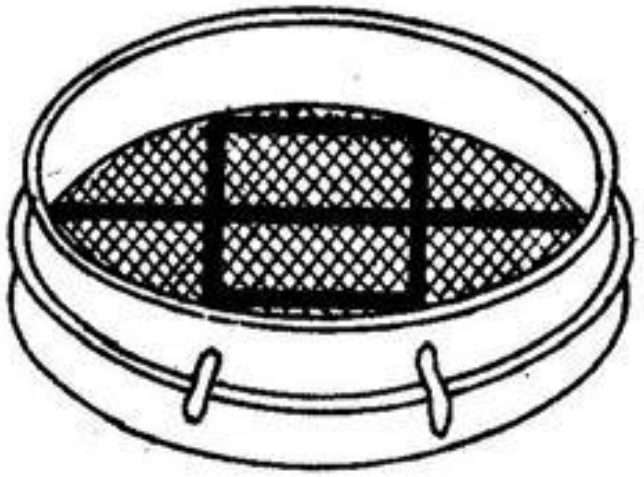
Trowels

Lifter: A lifter is a finishing tool that is used for repairing the mould and finishing the mould sand.

Lifter is also used for removing loose sand from mould.

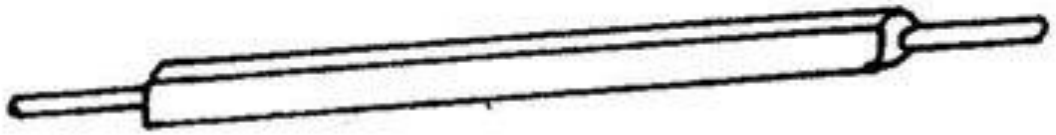
LIFTER

Hand riddle: It is used for ridding of sand to remove foreign material from it. It consists of a wooden frame fitted with a screen of standard **wire mesh** at the bottom.



Riddle

Strike off bar: It is a flat bar, made of wood or iron to strike off the excess sand from the top of a box after ramming.
Its one edge made beveled and the surface perfectly smooth and plane.

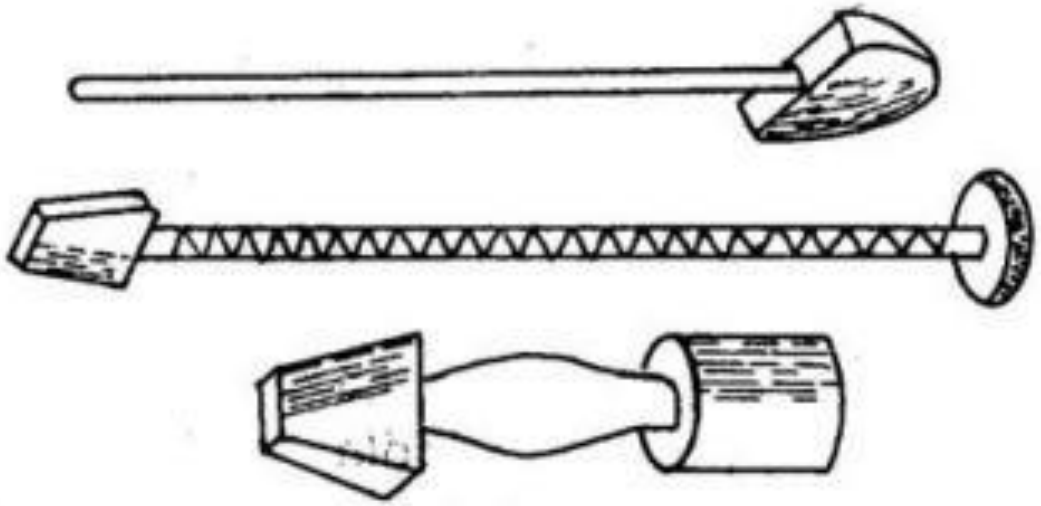


A strike off bar

Vent wire: It is a thin steel rod or wire carrying a pointed edge at one end and a wooden handle at the other. After ramming and striking off the excess sand it is used to make small holes, called vents, in the sand mould to allow the exit of gases and steam during casting.

Vent wire

Rammers: Rammers are used for striking the sand mass in the moulding box to pack it closely around one pattern. Common types of rammers are shown as under.



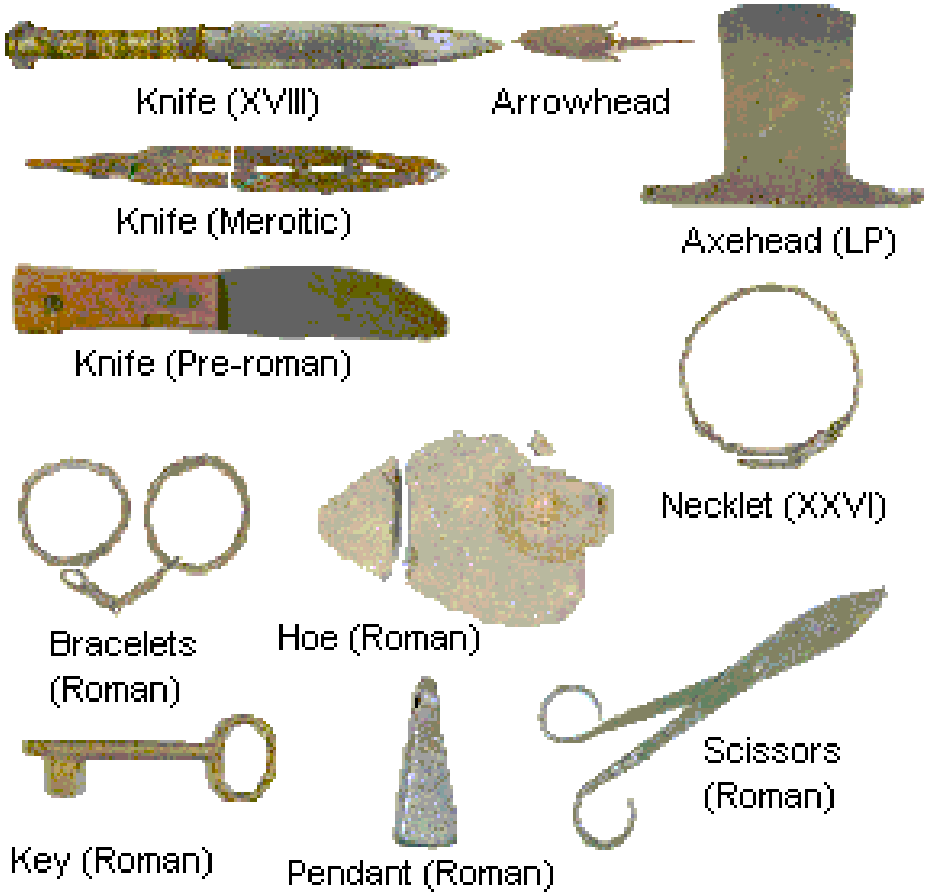
Rammers

Moulders Flask
Moulding Board
Sprinkler
Shovel
Rammer
HANDLE
BLADE
MOULDER' TROWEL
Moulder's Trowel
Vent Wire
Bulb Swab
Slicks & Spoons
Strike Off Bar
Gate Cutter
Ladles
Lifers
Corner Tools
Slicks, Spoons, Lifters and Cornering Tool

Casting

- **Oldest Manufacturing Process**

According to biblical records, casting dates back ~ 5000 years B.C (for Arrow heads, weapons e.t.c)



Casting

- Oldest Manufacturing Process

3200 B.C, A copper frog (oldest known casting in existence) was cast in Mesopotamia.

460 B.C, Bronze statue of Zeus was cast in Greece…

Casting

Iron Pillar, Quthub Minar, Delhi.

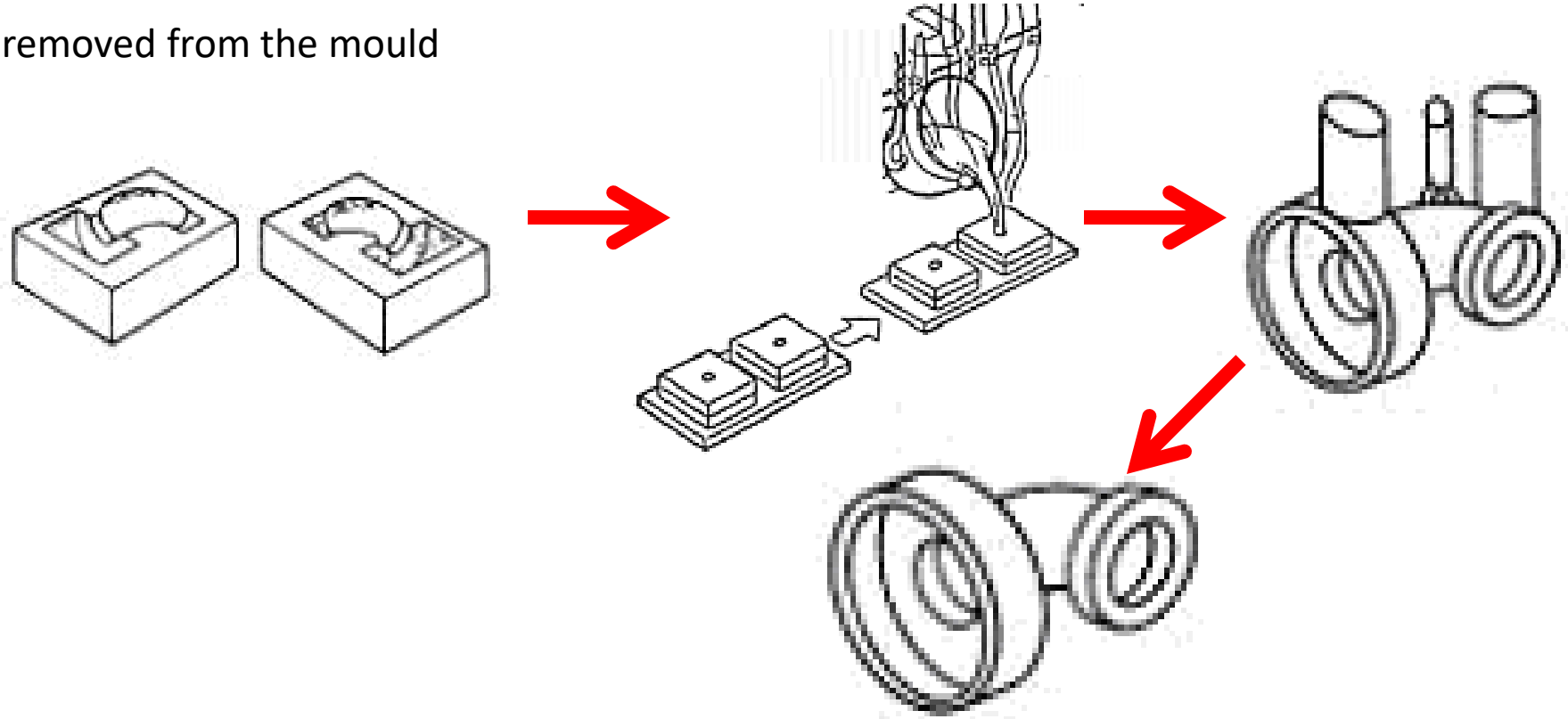
Iron Pillar ~ 400 A.D.

This iron pillar dating to 400 A.D., remains standing today in Delhi, India. Corrosion to the pillar has been minimal, and this skill is lost to current ironworkers.

Casting

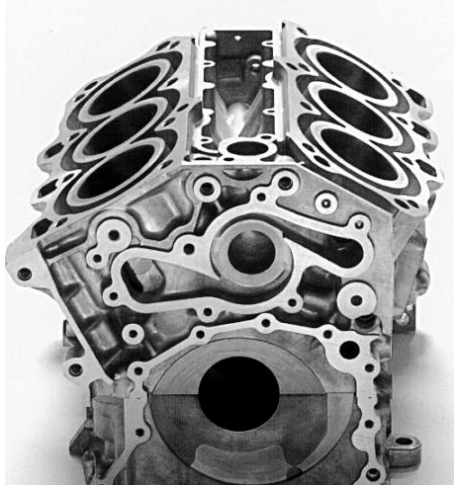
Principle of the Process

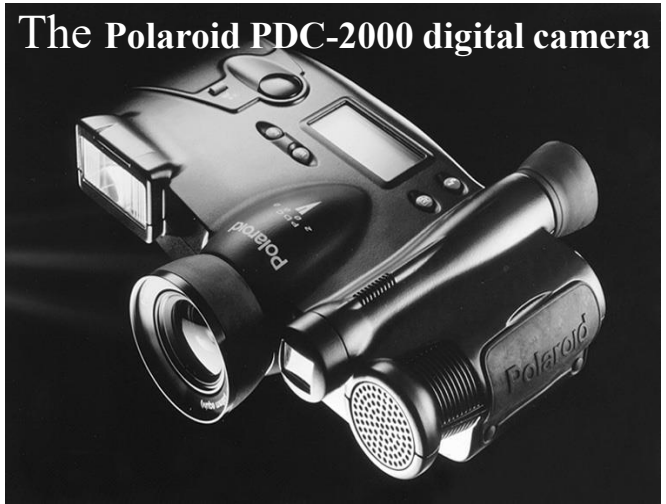
- Mould with required cavity is created
- Metal is heated above its melting temperature
- Liquid metal is poured into mould
- Metal solidifies inside the cavity of the mould (casting)
- Casting is removed from the mould



Casting

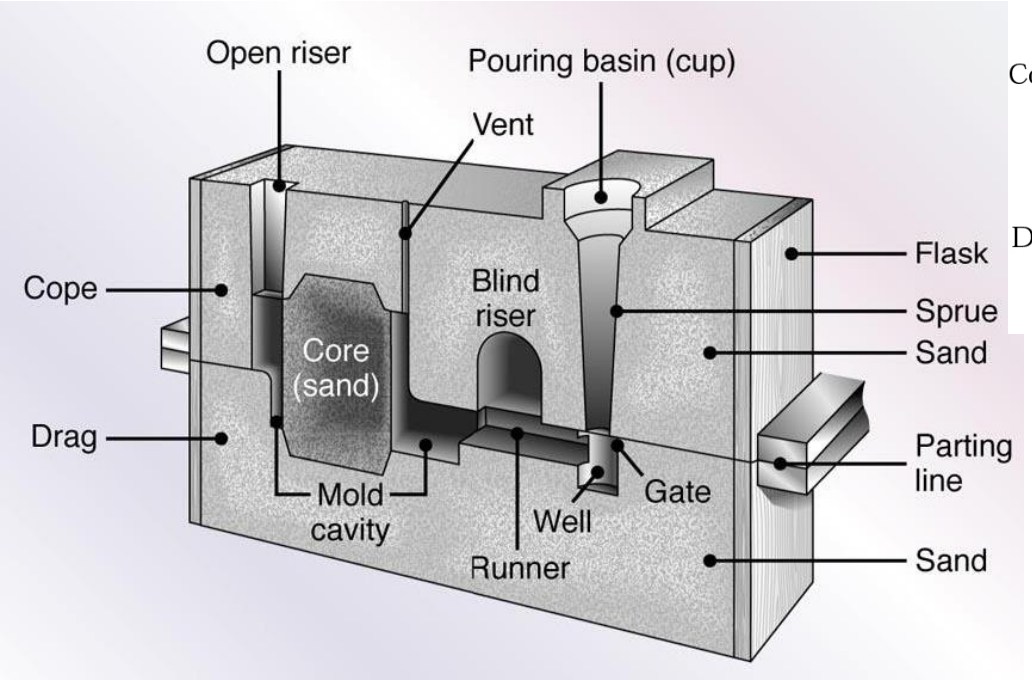
Examples of Castings

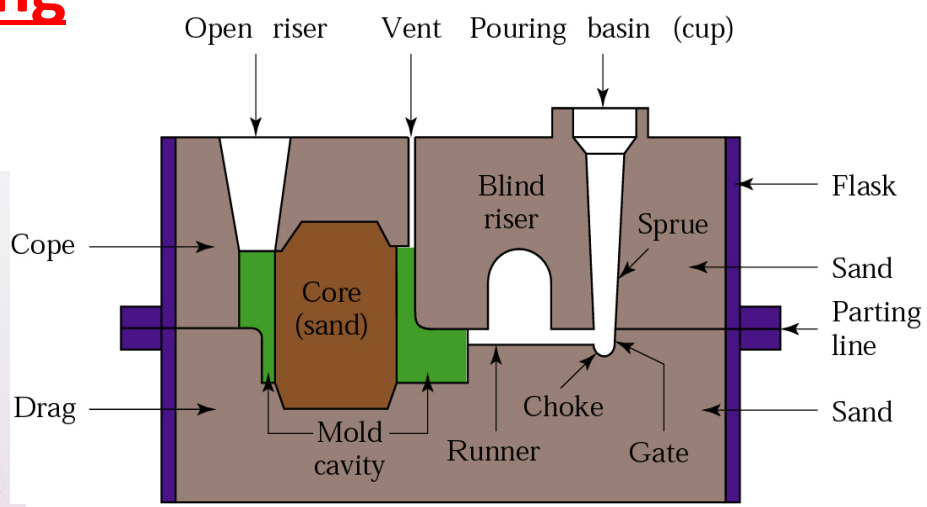




Casting Terminology

Casting





Schematic illustration of a sand mould, showing various features

Source: Kalpakjian & Schimd

Flask: **A metal or a wooden frame with out top or a bottom, in which mould is made.**

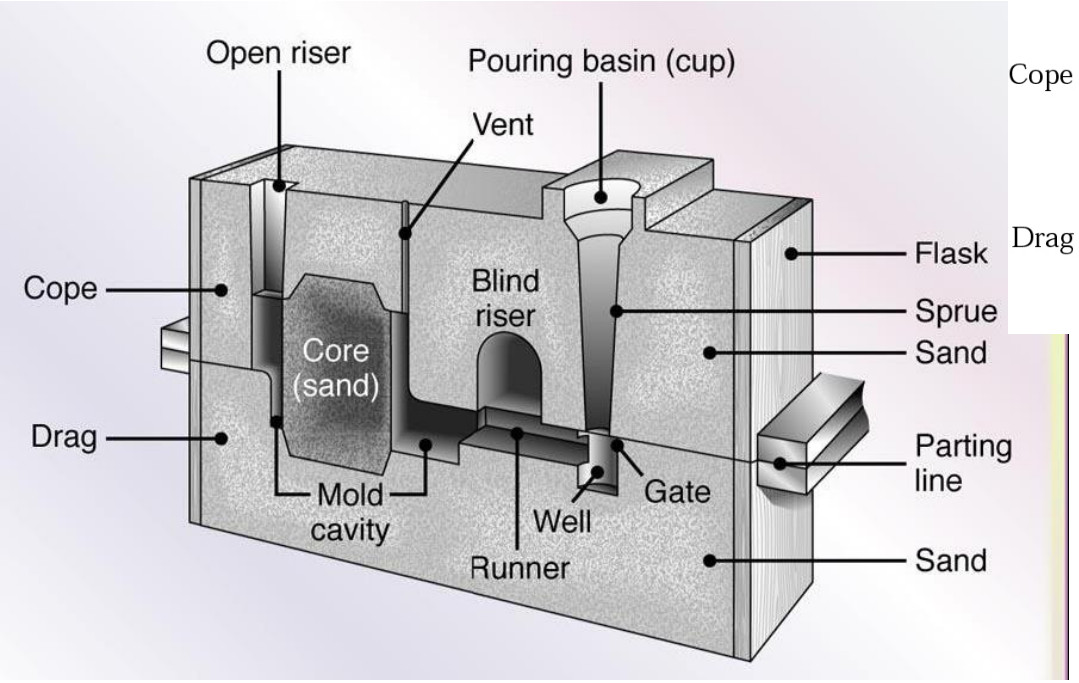
Cope: **Upper moulding flask**

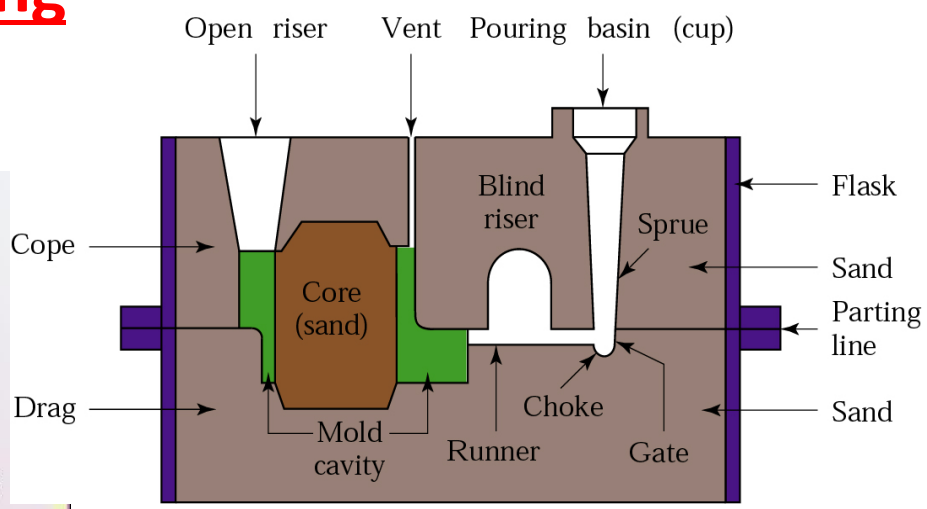
Cheek: **Intermediate moulding flask**

Drag: **Lower moulding flask**

Casting Terminology

Casting





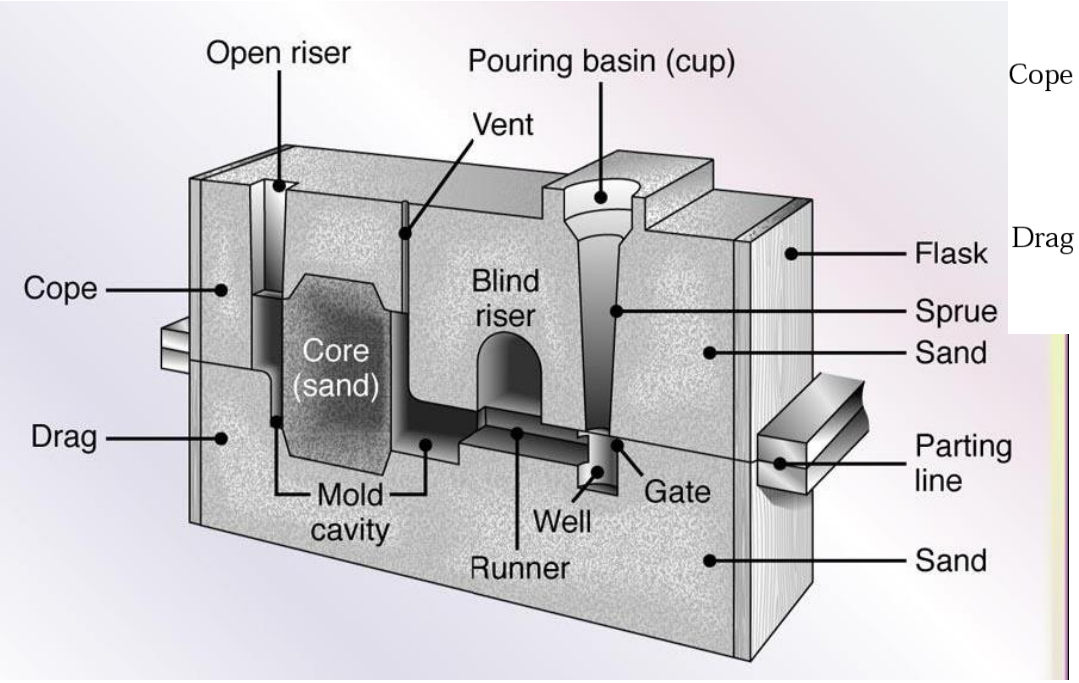
Schematic illustration of a sand mould, showing various features

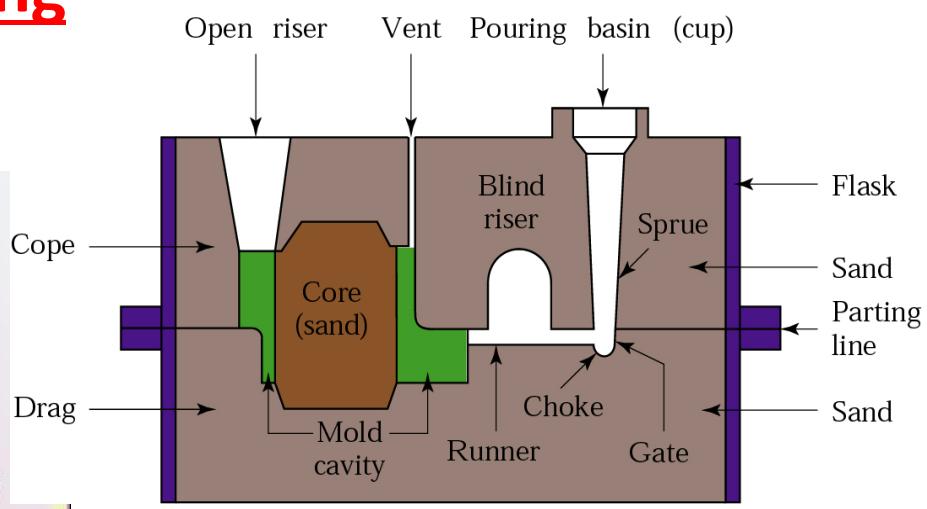
Source: Kalpakjian & Schimd

Pattern: **A physical model or the replica of the final object to be casted. Mould cavity is created with the help of the pattern.**

Casting Terminology

Casting





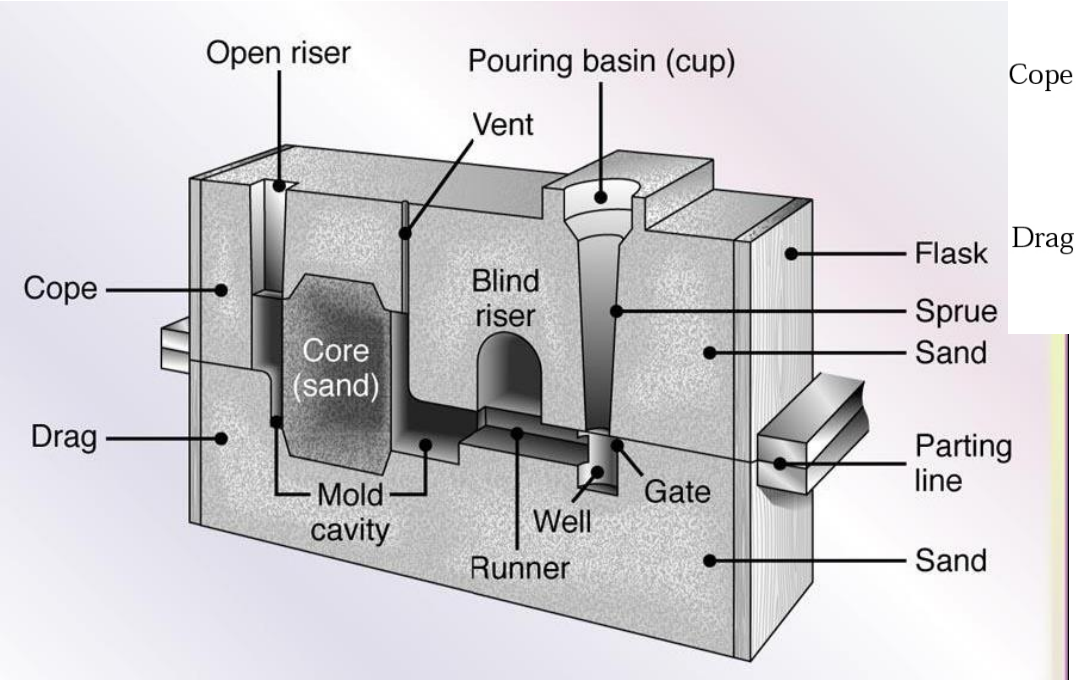
Schematic illustration of a sand mould, showing various features

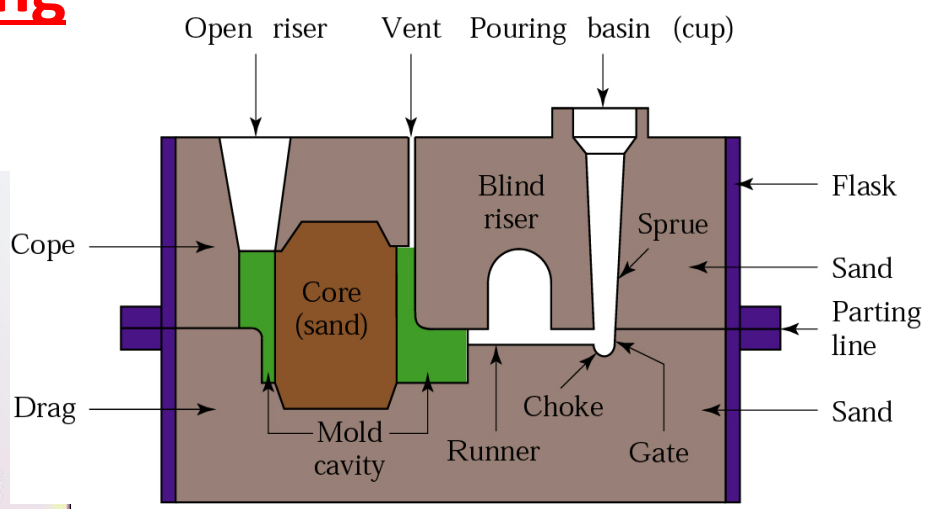
Source: Kalpakjian & Schimd

Parting Line: **A imaginary line that divides the drag & cope (two parts of the moulding flask) .**

Casting Terminology

Casting





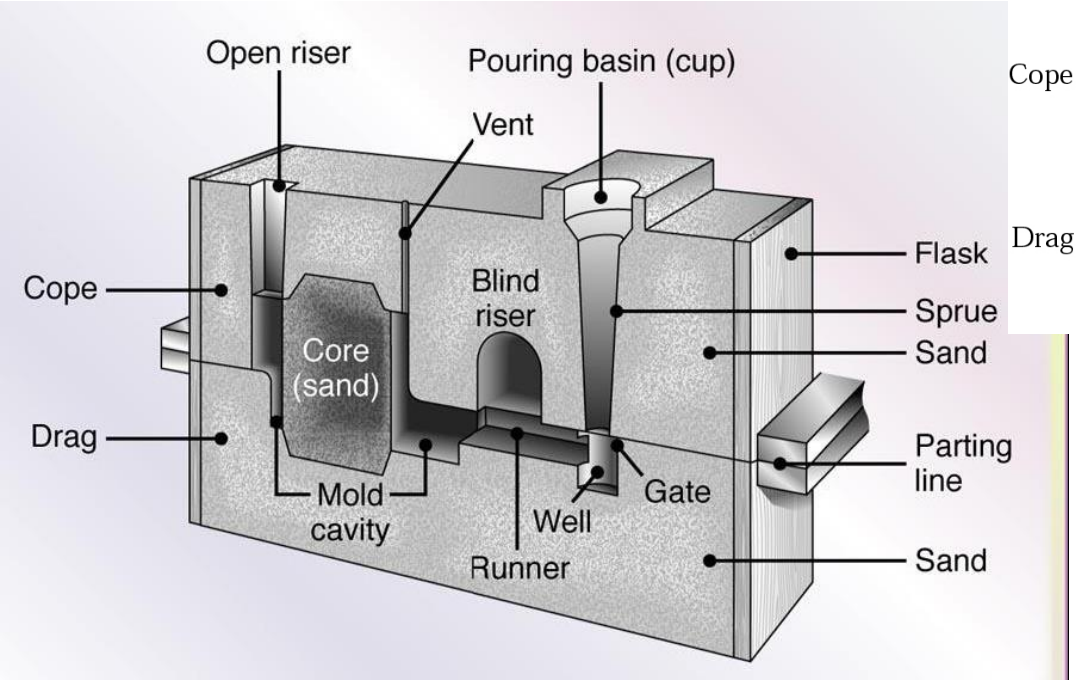
Schematic illustration of a sand mould, showing various features

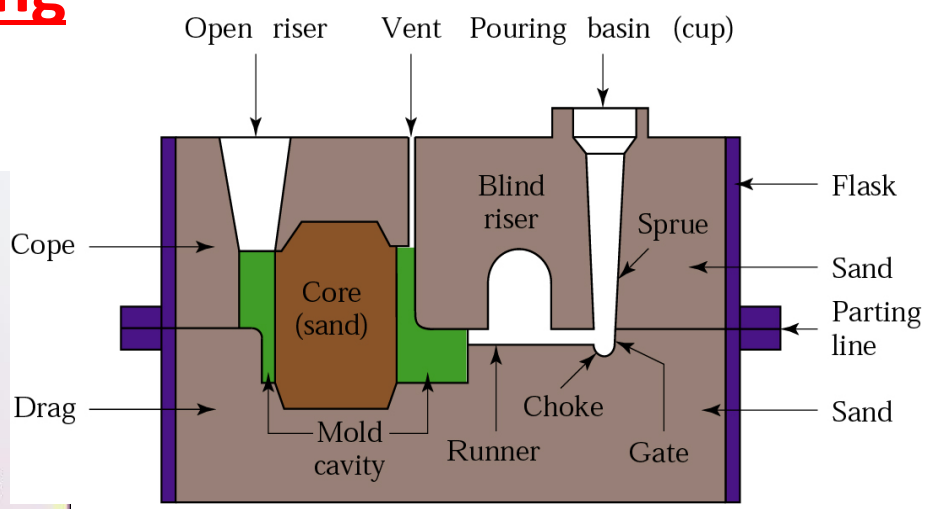
Source: Kalpakjian & Schimd

Moulding Sand: **Binding sand which is used to make the mould . It is the mixture of silica sand, clay and moisture in appropriate proportions and it is not supposed to loose permeability.**

Casting Terminology

Casting





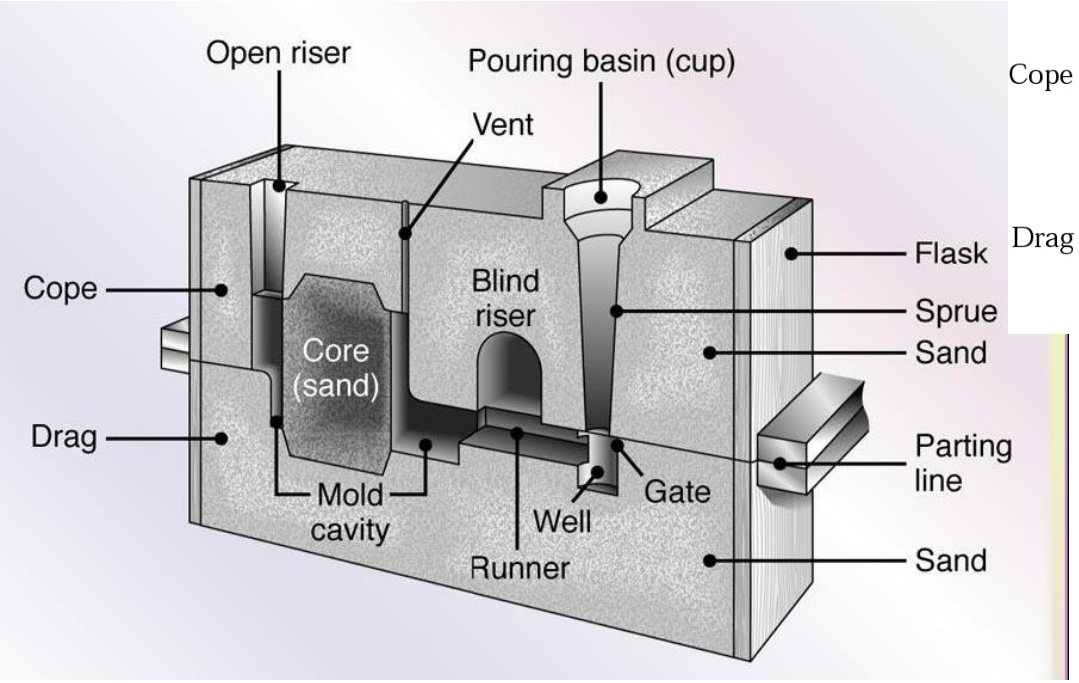
Schematic illustration of a sand mould, showing various features

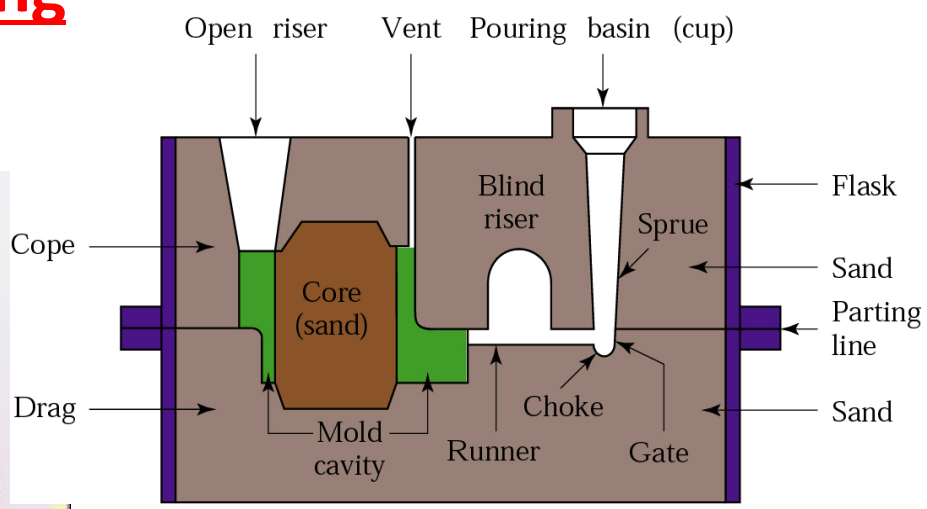
Source: Kalpakjian & Schimd

Core: **A separate part of the mould made of sand (conventionally baked), which is used to make various internal cavities inside the castings.**

Casting Terminology

Casting





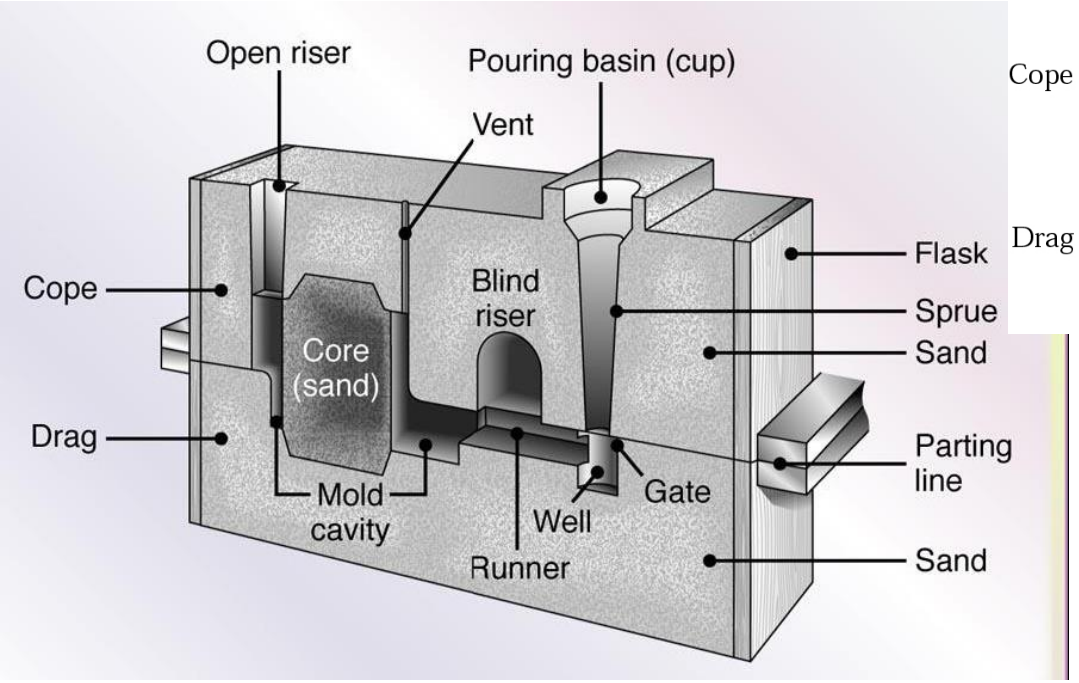
Schematic illustration of a sand mould, showing various features

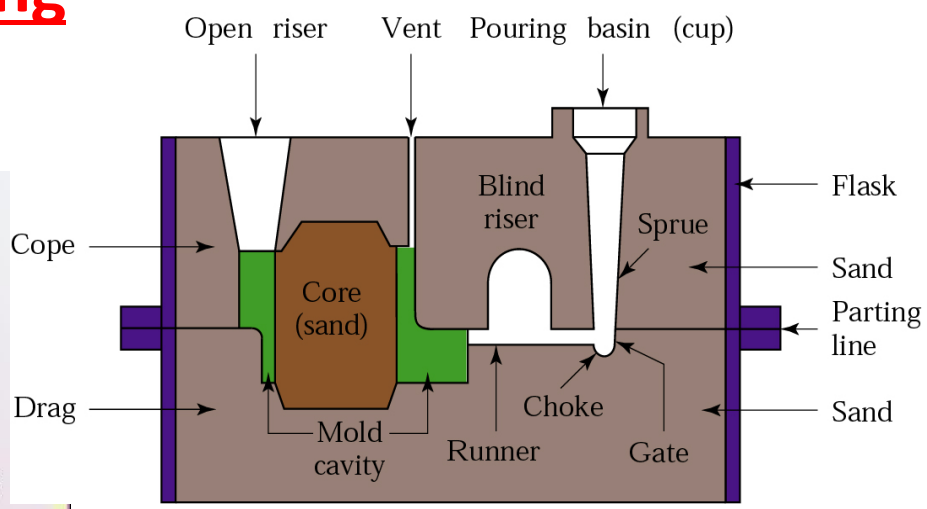
Source: Kalpakjian & Schimd

Pouring Basin: **A small funnel shaped cavity at the top of the mould into which molten metal is poured.**

Casting Terminology

Casting





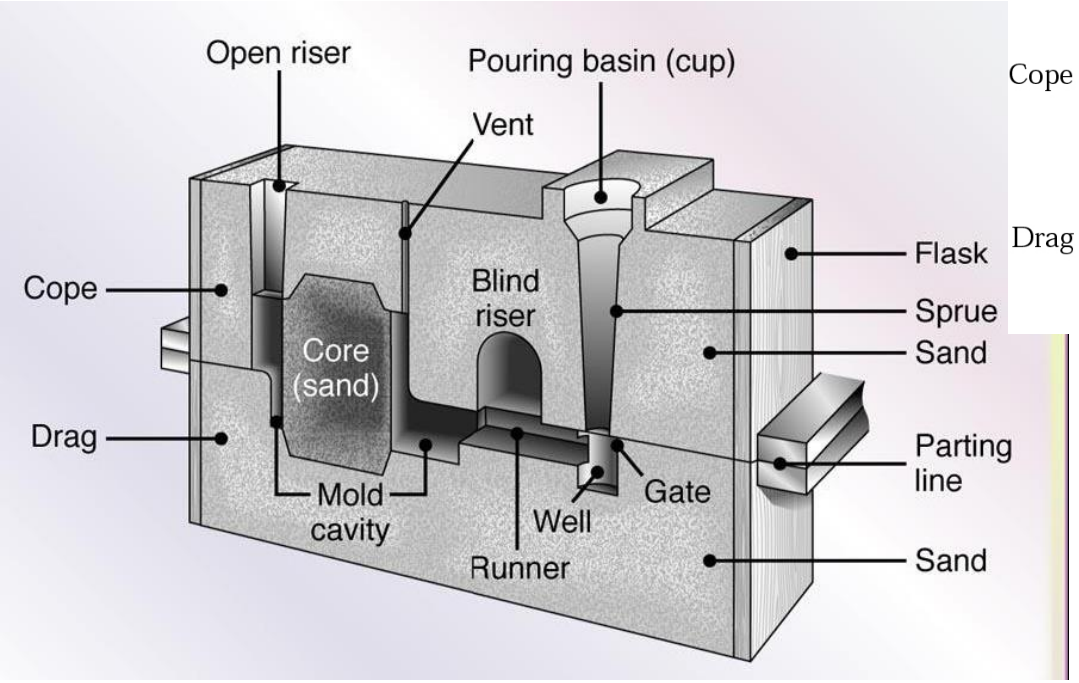
Schematic illustration of a sand mould, showing various features

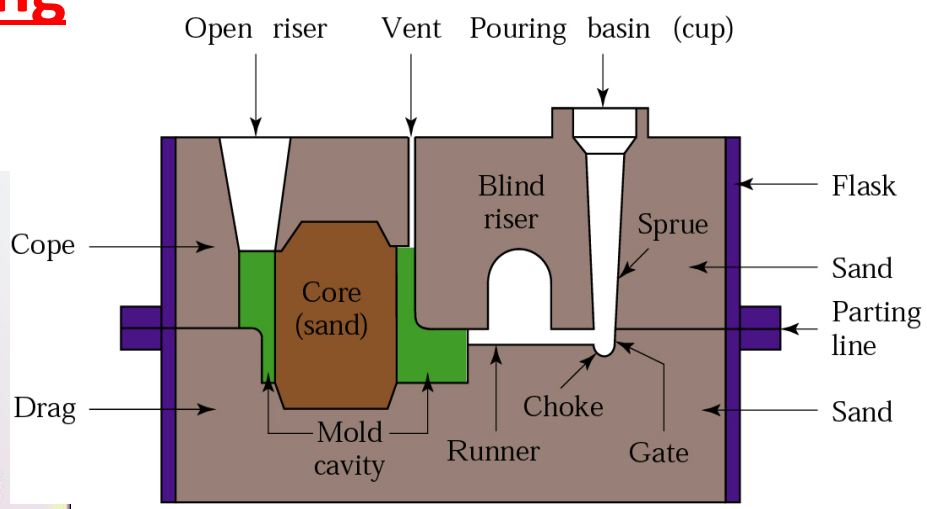
Source: Kalpak Jian & Schimd

Sprue: **The passage through which the molten metal flows from pouring basin to mould cavity.**

Casting Terminology

Casting





Schematic illustration of a sand mould, showing various features

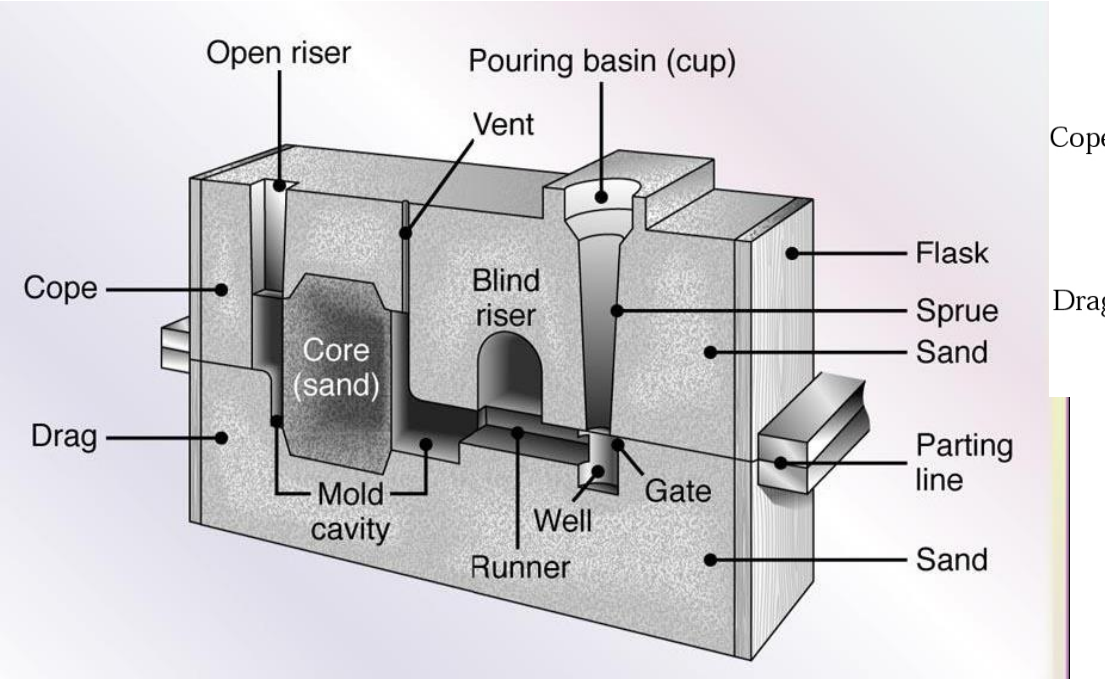
Source: Kalpakjian & Schimd

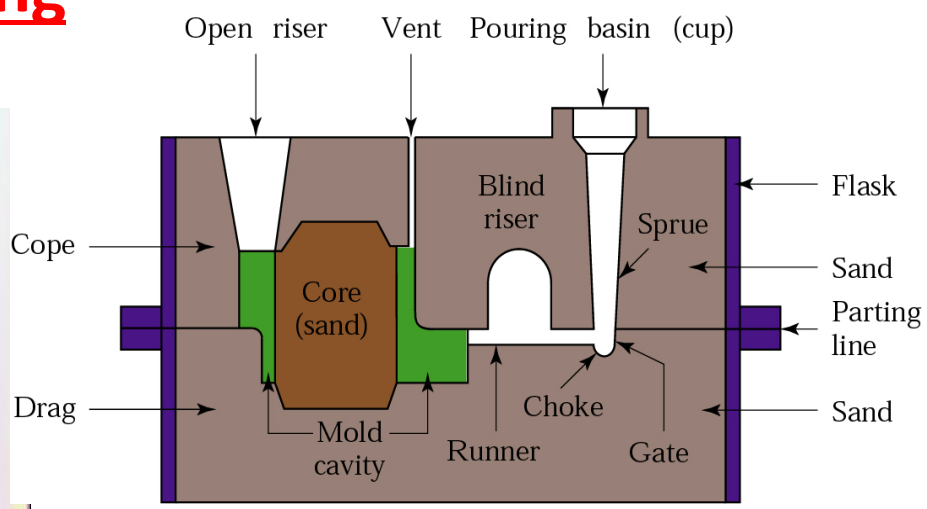
Gate: **The channel through which the molten metal enters the mould cavity.**

Runner: **The channel through which the molten metal is carried from Sprue to Gate.**

Casting

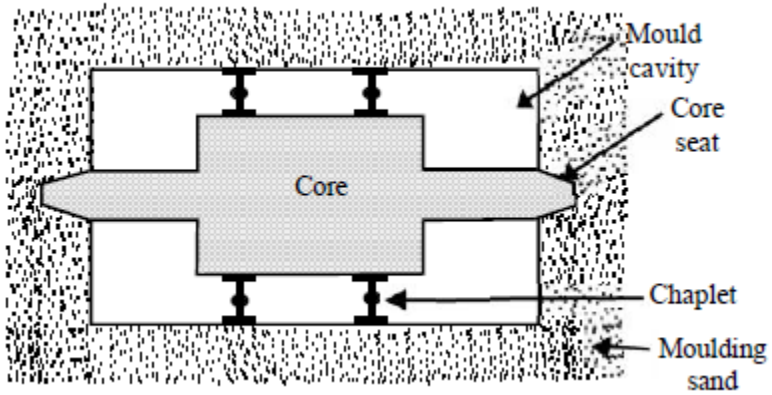
Casting Terminology





Schematic illustration of a sand mould, showing various features

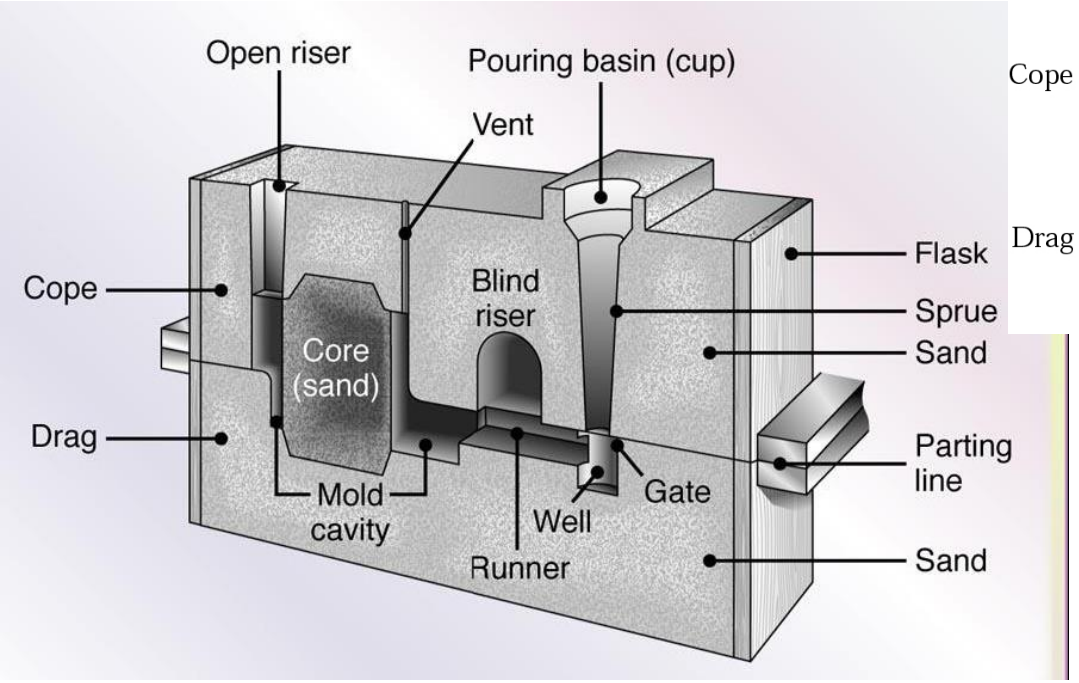
Source: Kalpakjian & Schimd

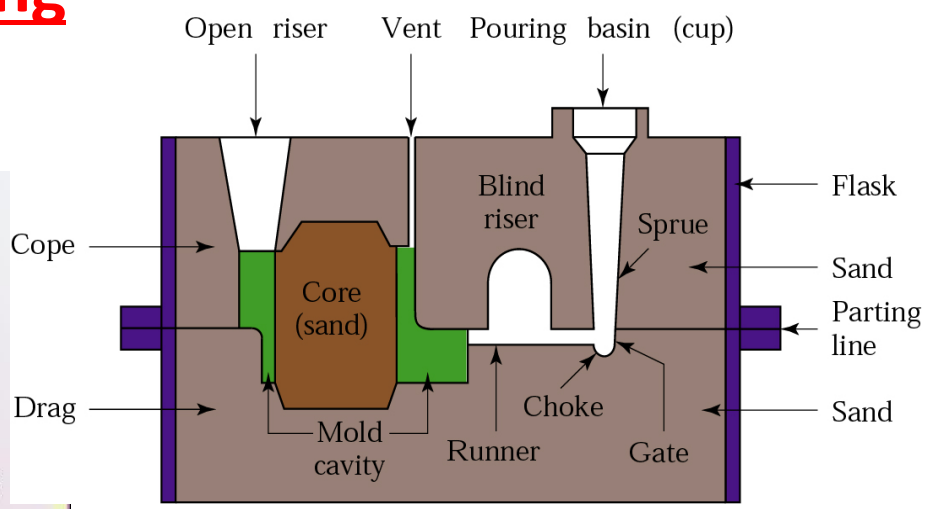


Chaplets: **The metallic supports used to held the Core inside the mould cavity, to with stand its own weight and resist metallostatic forces.**

Casting Terminology

Casting





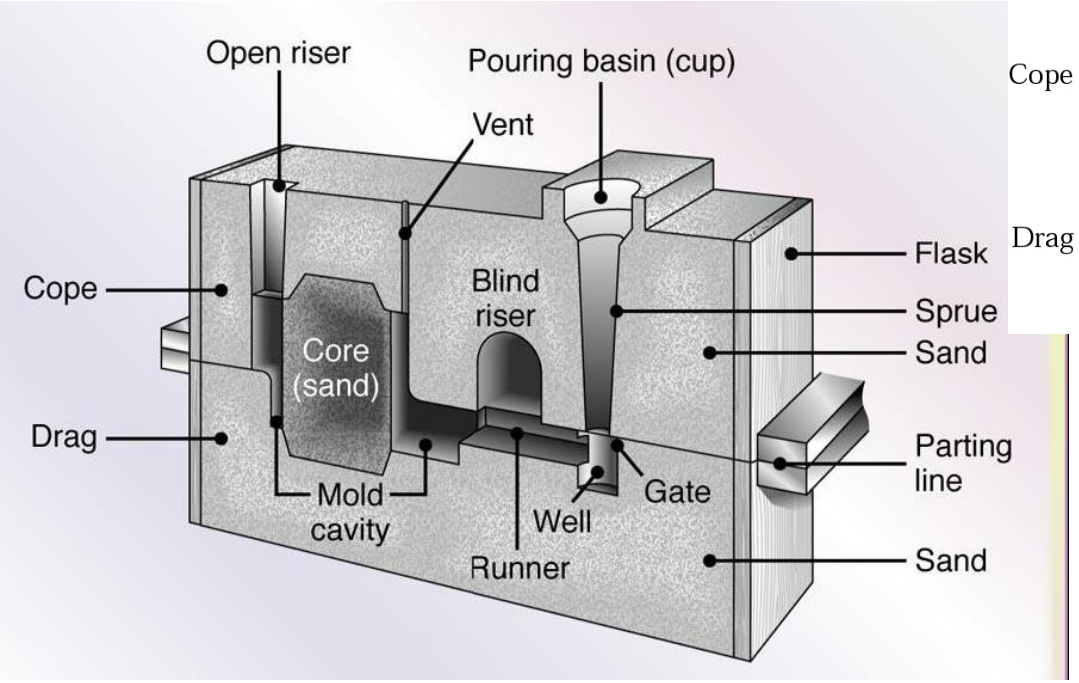
Schematic illustration of a sand mould, showing various features

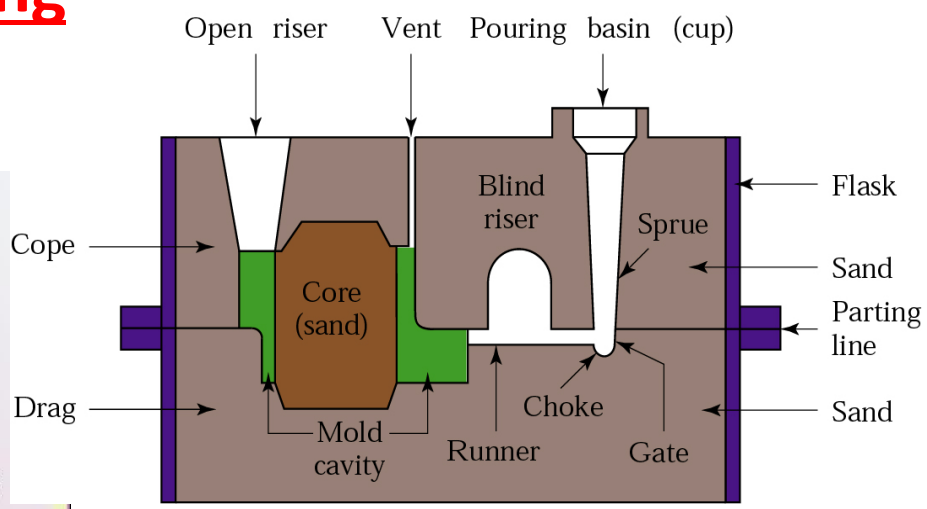
Source: Kalpakjian & Schimd

Riser: **The extra void created in the mould that will be filled by the molten material. It simple functions as a reservoir of molten metal for the castings, to compensate material shrinkage , which occurs during solidification.**

Casting Terminology

Casting





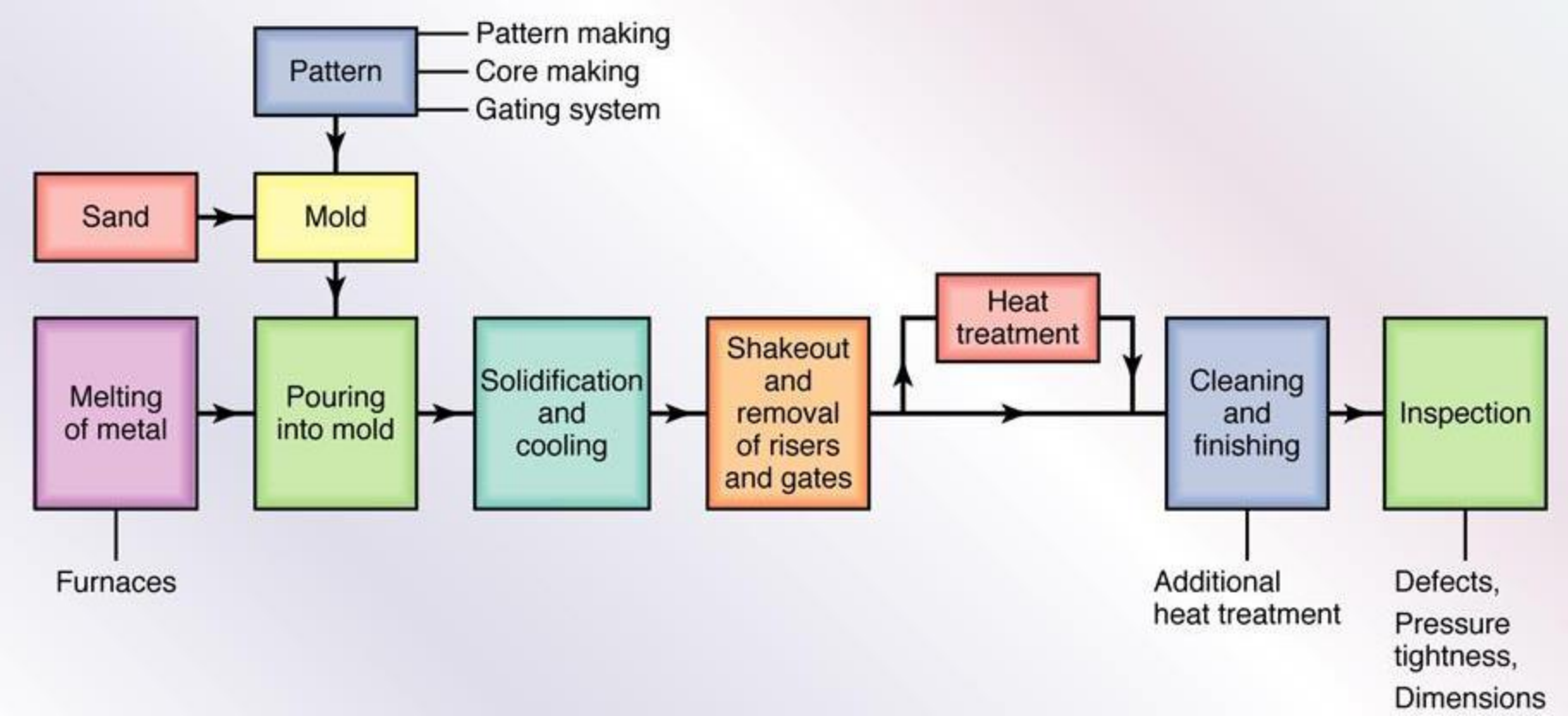
Schematic illustration of a sand mould, showing various features

Source: Kalpakjian & Schimd

Vent: **Small opening provided in the mould to facilitate escape of air (from the mould) and the gases (from the molten metal).**

Casting

Casting Procedure



Outline of production steps in a typical sand-casting operation

Source: Kalpakjian & Schimd

Casting

Advantages

- Very complicated shapes (Close tolerances),
- Internal cavities creation
- Very large castings,
- Very small castings,
- Use is widespread; technology well developed and more diverse range of products.
- Materials are inexpensive (sand)
- Process is suitable for both ferrous, non-ferrous metal and alloys castings,
- Cast metal is isotropic, It has same physical and mechanical properties on all directions.

Casting

Disadvantages

- Material waste often ~20 – 50 %
- Slow production rate
- Labor intense job,
- Rough surface finish,
- Sand **:** metal ratio is relatively high.
- High level of waste is typically generated, particularly sand, refractory linings, bag house dust, spent shot, timber (from pallets and pattern making), e.t.c.

Patterns and Pattern Making - Casting

Pattern

Pattern is a principle tool in casting process. It is replica/physical model of the final object to be casted, with some small variations.

Major modifications include:

- Pattern Allowances,
- Provision of core.

Functions of Pattern:

- It helps in preparation of the mould,
- It enables creation of core points,
- It makes provision for runner, gates and riser,
- Patterns properly made and having smooth surfaces reduce further finishing and reduces casting defects.,
- Properly constructed castings reduce the overall cost of the casting.

Properties of Sand - Casting

Composition :

- **70-85% Silica sand (SiO_2)**
- **10-12% Bonding material e.g., clay etc.**
- **3 – 6 % Water**

Base Sand :

Silica sand is most commonly used base sand. Other base sands that are also used for making mold are zircon sand, Chromite sand, and olivine sand. Silica sand is cheapest among all types of base sand and it is easily available.

Properties of Sand - Casting

Binder :

Binders are of many types such as:

1. Clay binders,
2. Organic binders and
3. Inorganic binders

Clay binders are most commonly used binding agents mixed with the molding sands to provide the strength. The most popular clay types are:

Kaolinite or fire clay ($Al_2O_3.2SiO_2.2H_2O$) and Bentonite ($Al_2O_3.4SiO_2\ nH_2O$)

Of the two, the Bentonite can absorb more water which increases its bonding power.

Moisture :

Clay acquires its bonding action only in the presence of the required amount of moisture. When water is added to clay, it penetrates the mixture and forms a microfilm, which coats the surface of each flake of the clay. The amount of water used should be properly controlled. This is because a part of the water, which coats the surface of the clay flakes, helps in bonding, while the remainder helps in improving the plasticity.

Properties of Sand - Casting

Composition :

- **70-85% Silica sand (SiO_2)**
- **10-12% Bonding material e.g., clay etc.**
- **3 – 6 % Water**

Requirements :

- Refractoriness – Ability to remain solid at high temp
- Cohesiveness – Bonding
- Permeability – Gas flow through mould
- Collapsibility – Ability to permit metal to shrink after solidification

Criteria :

- Permeability
- Green strength
- Dry strength

Properties of Sand- Casting

Flowability:

It is the ability of the moulding sand to flow and get compacted all around the pattern and take up the required shape.

Refractoriness:

It is the ability of the moulding sand to withstand the high temperature of the molten metal which is to be poured. Silica sand has the high refractoriness…

Permeability:

It is the ability of the moulding material to allow hot gasses to pass through it…

Properties of sand- Casting

Green Strength:

The moulding sand that contains the moisture is termed as green sand. Green strength is the ability of the green sand to retain the shape of the construct the model.

Dry Strength:

It is the ability of the moulding material to retain the exact shape of the mould cavity in the dry condition (when the molten metal is poured in the mould) and to withstand the metallostatic forces of the molten metal.

Hot Strength:

It is the ability of the moulding material to retain the exact shape of the mould cavity at an elevated temperature.

Elements of Gating System

Main Elements :

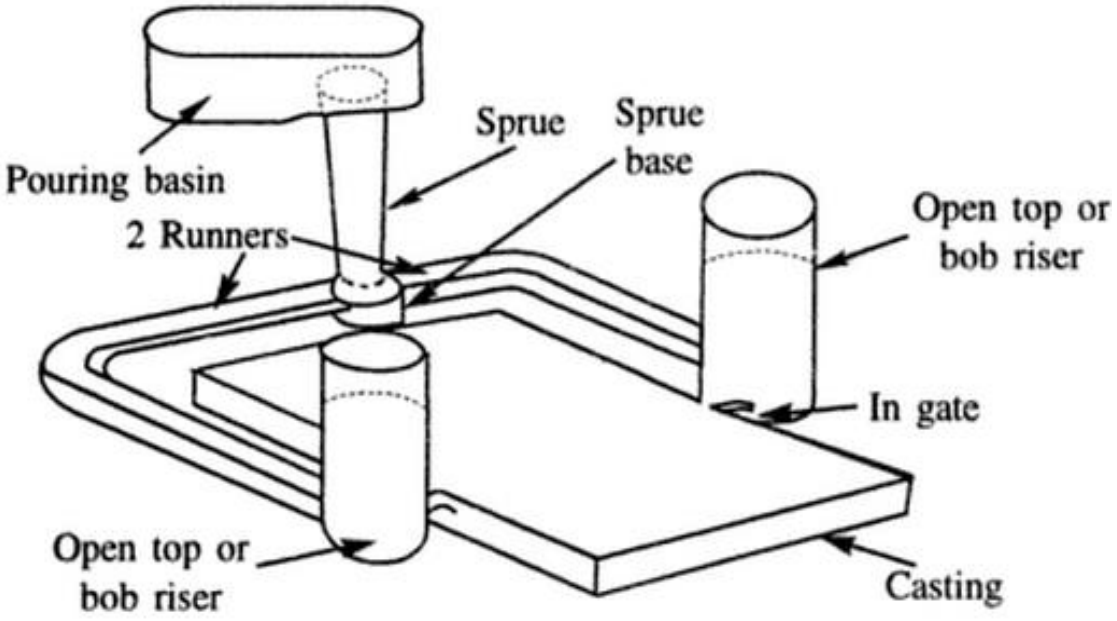


Fig. 10.1 Typical gating system

- pouring basin
- sprue
- sprue base well
- runner
- runner extension
- ingate
- riser

Elements of Gating System

Functions :

1. The mould should be completely filled in the smallest time possible without having to raise metal temperature nor use higher metal heads.
2. The metal should flow smoothly into the mould without any turbulence. A turbulent metal flow tends to form dross in the mould.
3. Unwanted material such as slag, dross and other mould material should not be allowed to enter the mould cavity.
4. The metal entry into the mould cavity should be properly controlled in such a way that aspiration of the atmospheric air is prevented.
5. A proper thermal gradient should be maintained so that the casting is cooled without any shrinkage cavities or distortions.
6. Metal flow should be maintained in such a way that no gating or mould erosion takes place.
7. The gating system should ensure that enough molten metal reaches the mould cavity.
8. The gating system design should be economical and easy to implement and remove after casting solidification.
9. Ultimately, the casting yield should be maximised.

CARPENTRY

INTRODUCTION

Carpentry may be defined as the process of making wooden goods and components, viz. roofs, floors, partitions, doors, etc., with the help of carpentry tools. It starts from a marketable form of wood and ends with a finished product. It deals with the specific work of a carpenter like making different types of joints to form a finished product.

Types of woods

Hardwoods: Oak, Beech, Ash, Walnut, Maple
Softwoods: Pine, Redwood

TIMBER

Timber is wood that has been processed into dimensional wood, including beams and planks or boards, a stage in the process of wood production.

The wood obtained from fully grown trees is cut and is prepared for engineering purposes. This wood is known as *timber*. Timber of a live tree is called *stationary timber*, felled trees after sawing is called *rough timber* and after cutting into two various sizes as *converted timber*.

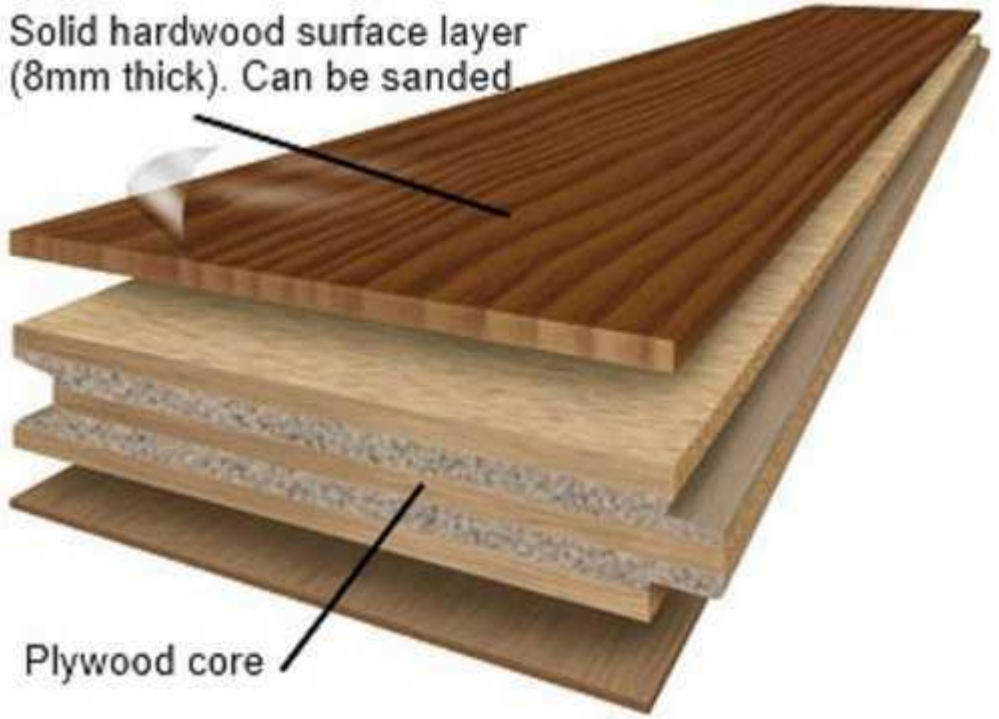
The word, 'grain', as applied to wood, refers to the appearance or pattern of the wood on the cut surfaces. The grain of the wood is a fibrous structure and to make it strong, the timber must be so cut, that the grains run parallel to the length.

Advantages of Timber

Timber has the following advantages over the other materials:

1. It is cheaper and easily available.
2. It is lighter and stronger than most of the materials used in construction work.
3. It is very easy to work on it by the help of different tools for getting required shapes.
4. Cost of construction and workmanship is low.
5. Structural construction and joints can be easily made in timber.
6. It has high salvage value.
7. It responds very well to polishing and painting, etc.
8. It is suitable for making soundproof constructions.
9. It is a nonconductor of heat and electricity.
10. It is very suitable for making doors, windows, cabinet work, decorative designs and fitting, etc.

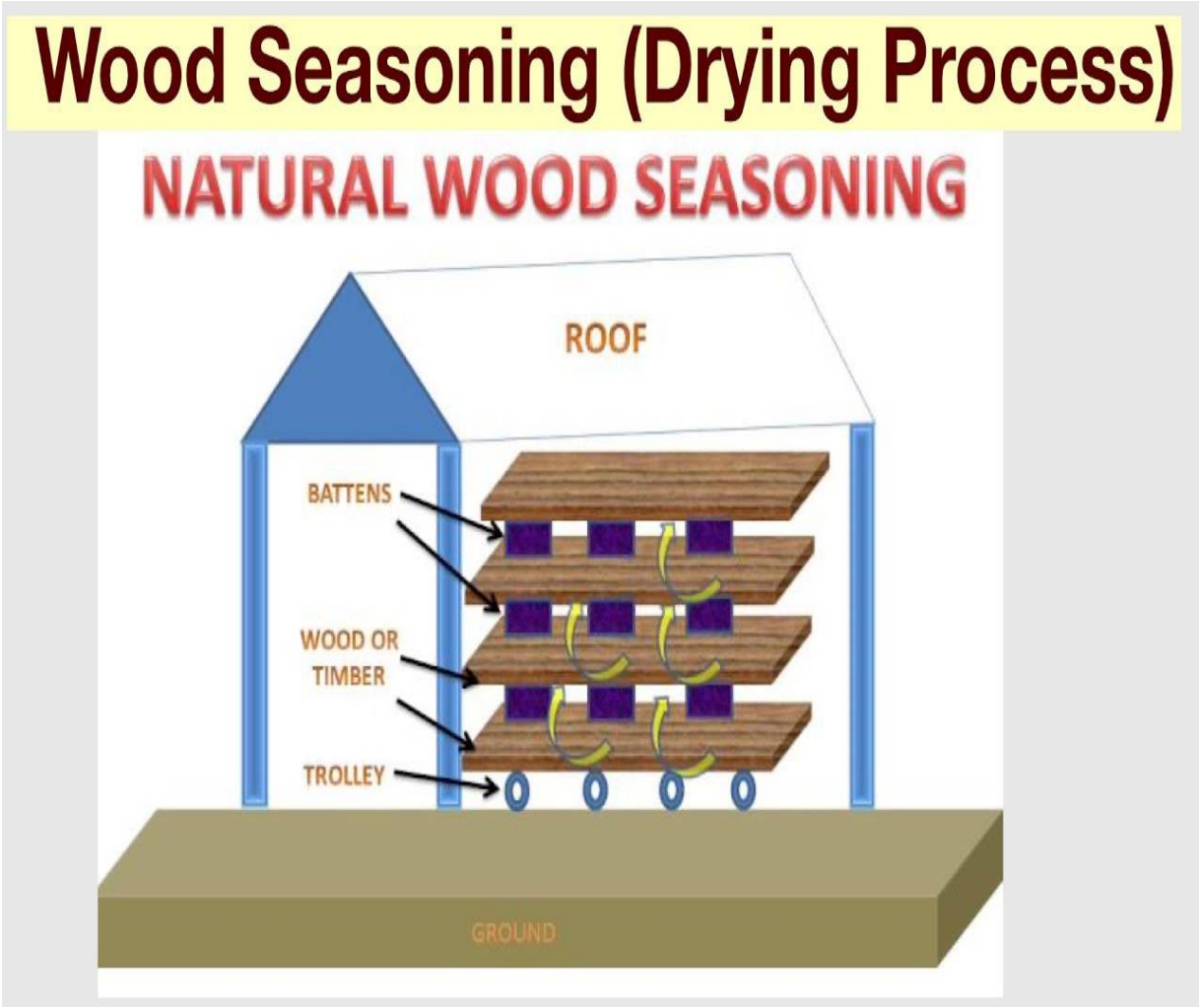
Engineered wood is wood that although it comes from wood, it's been processed to be something slightly different and perform in a different, oftentimes better way than what just the raw wood would perform



SEASONING OF WOOD

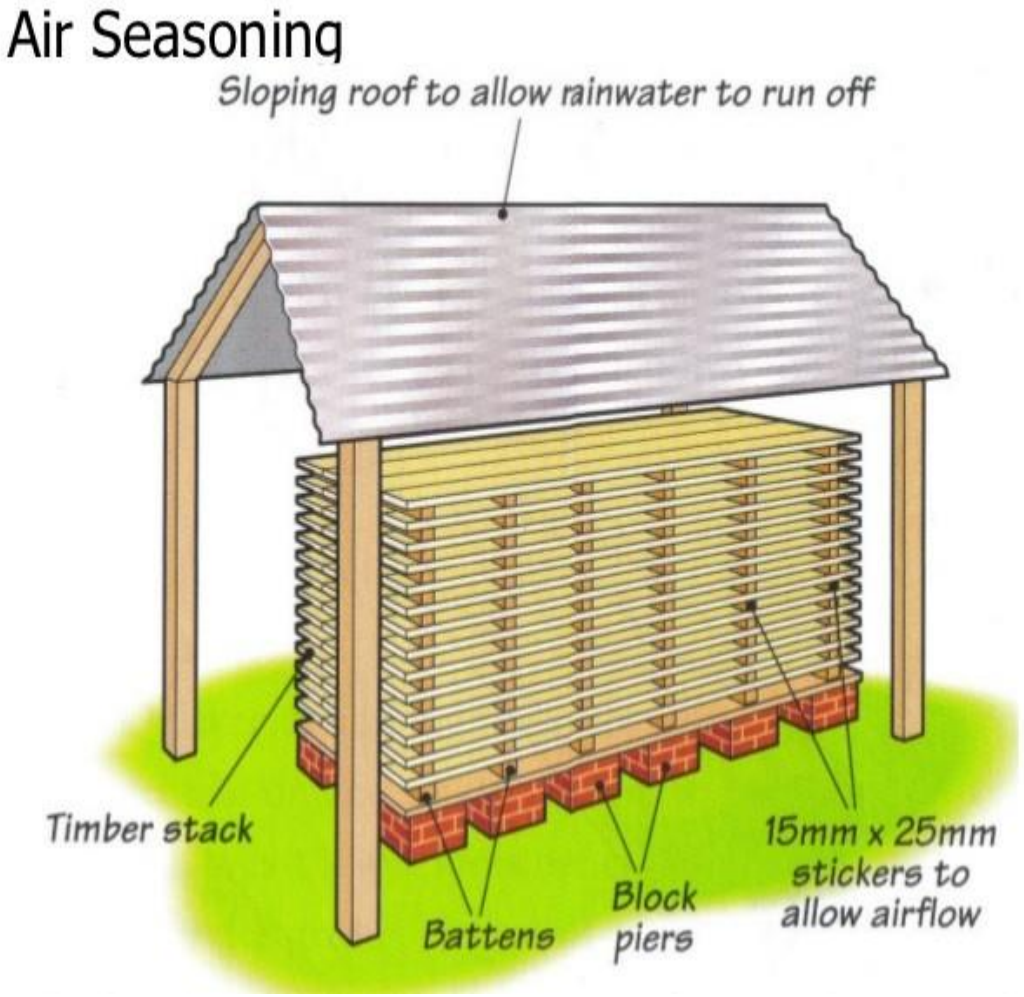
Seasoning is the process of drying timber to remove the **bound moisture** contained in walls of the wood cells to produce seasoned timber.

Seasoning can be achieved in a number of ways, but the aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying



Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting in service.

In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength.



Seasoning of
timber

Seasoning is the controlled process of reducing the moisture content of the timber such that it is suitable for the environment and intended use. It is very important to get this done before using it for construction purpose.

The advantages are as follows:

- Seasoned wood will be lighter, stronger and reliable.
- It is better against wood rot and other wooden parasites (wood worm).
- it will behave better against glue and adhesives
- It will not warp or shrink further once seasoned.

Natural Seasoning (Air) – In this process, timber is stacked in the open air, preferable at site. Stacking is done with horizontal distance of at least 25mm. The stack pile should be kept raised above the ground to provide good circulation of air and free from rising dampness. Lastly, overhead cover should be placed to prevent from effects of direct sunlight.

Softwood takes about 3 months, and hardwood about 12 months for seasoning.

Artificial Seasoning (Kiln) – In this process, the seasoning relies on controlled environment to dry out the timber.

It either uses hot air (air mixed with gases) or superheated steam with latter being more common. The time consumed in this is significantly less compared to traditional air seasoning. It is primarily used for large scale lumber seasoning.

Chemical Seasoning of Timber (Salt Seasoning).

Electrical Seasoning of Wood.

Seasoning by Boiling.

Characteristics of Good Timber

1. It should have minimum moisture content, i.e., the timber should be well seasoned.
2. The grains of wood should be straight and long.
3. It must retain its straightness after seasoning.
4. It should produce near metallic sound on hammering.
5. It should be free from knots or cracks.
6. It should be of uniform color, throughout the part of the wood.
7. It should respond well to the finishing and polishing operations.
8. During driving the nails and screws, it should not split easily.

CARPENTRY TOOLS

1. Marking and Measuring Tools

2. Holding Tools

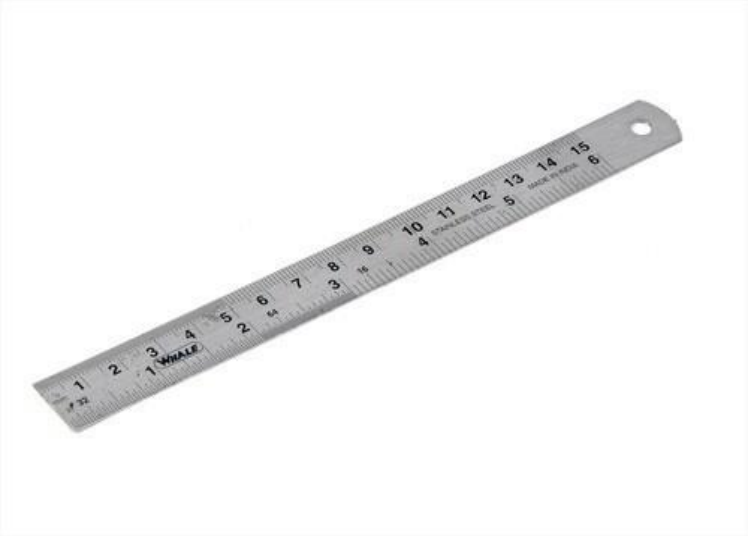
3. Planning Tools

4. Cutting Tools

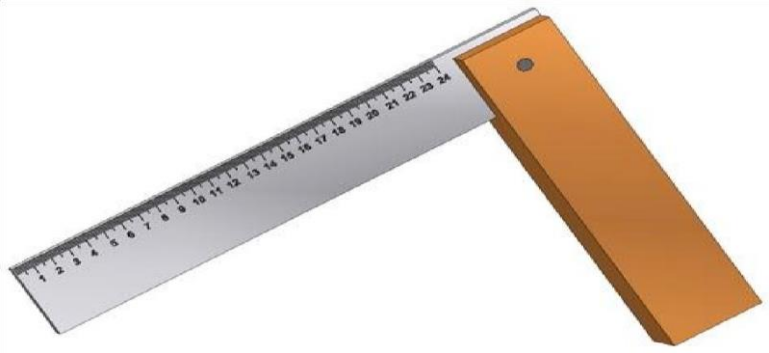
5. Drilling and Boring Tools

6. Miscellaneous Tools

Marking and Measuring Tools



Inch Tape &Steel rule

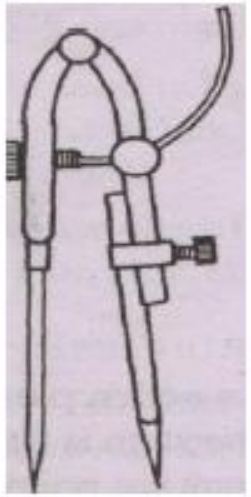


Try Square

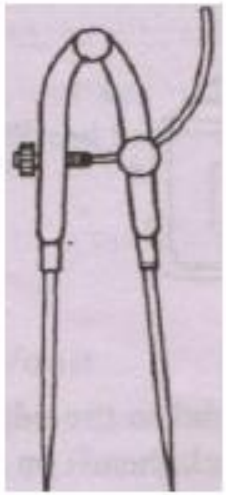
MARKING GAUGE

MORTISE GAUGE

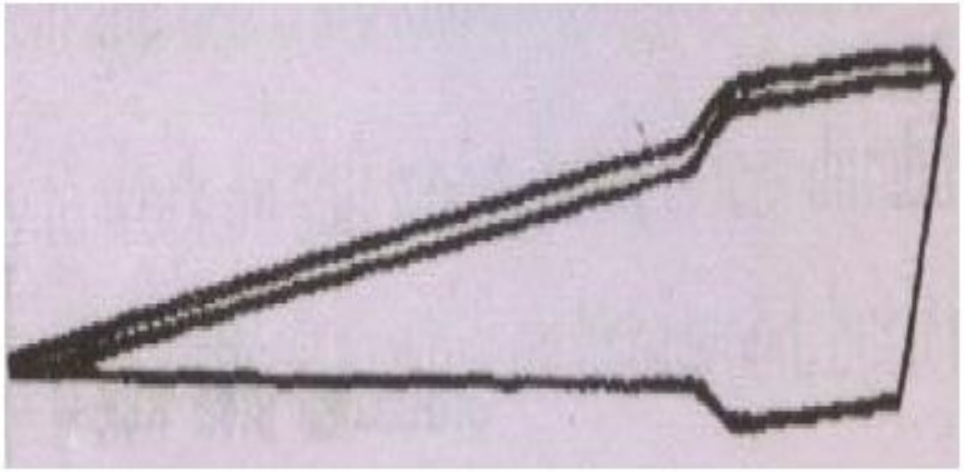
A mortise gauge or mortice gauge is a woodworking tool used by a carpenter or joiner to scribe mortise and tenon joints on wood prior to cutting



Compass



Divider

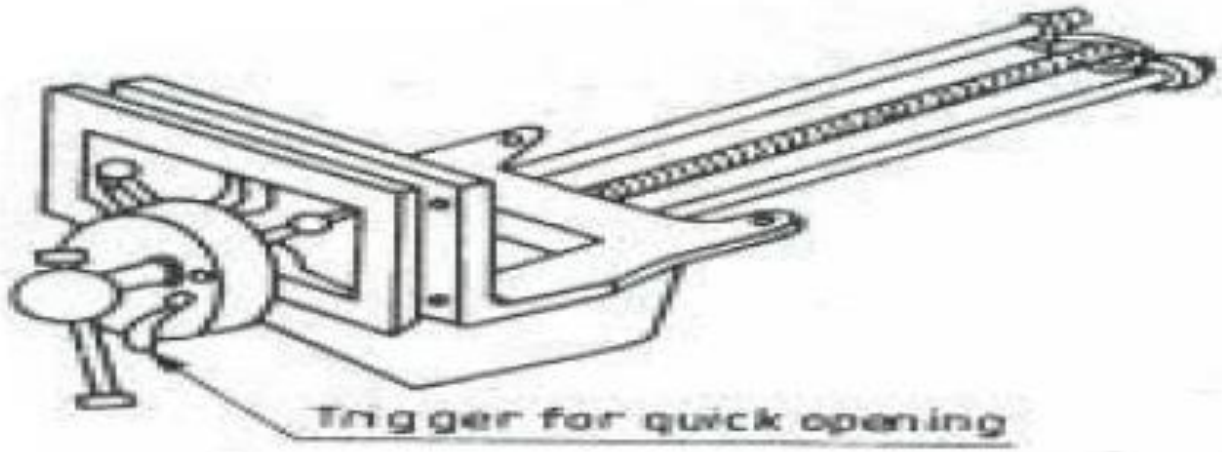


Scriber or marking knife

Bevel carpentry involves making angled cuts, or bevels, in wood to create an angled edge

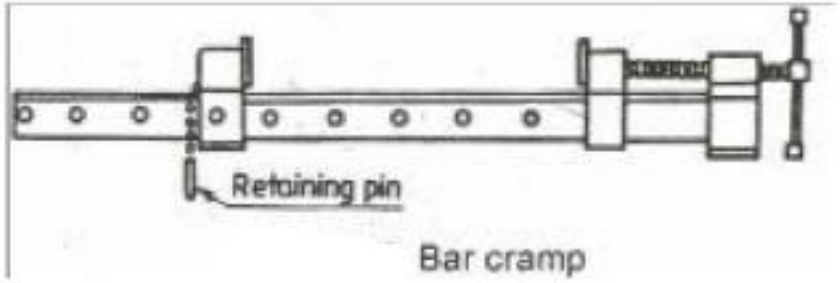
Bevel

Holding Tools



Carpenter's vice

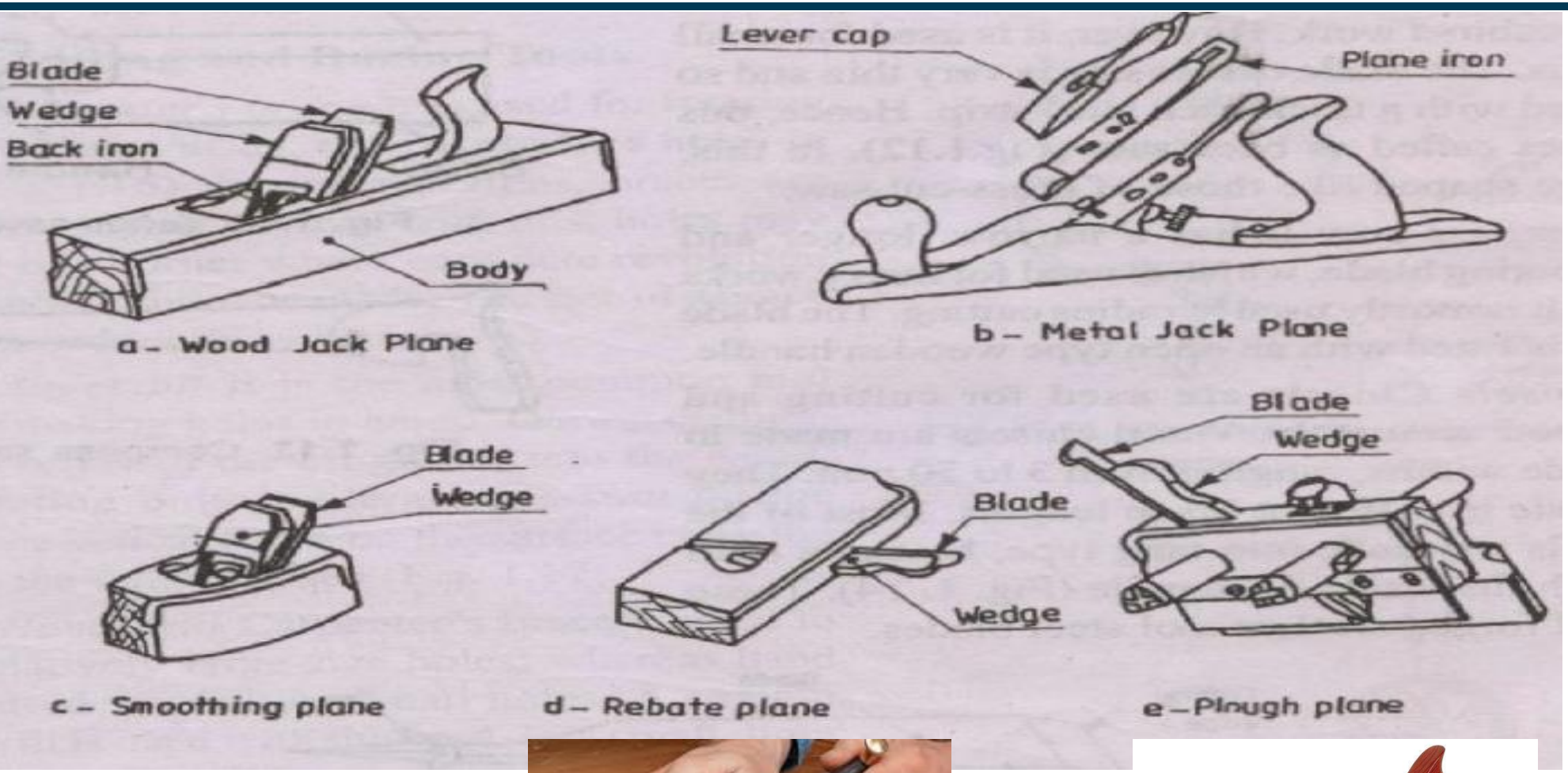
C-clamp



Bar cramp

Planning Tools

Types of Planes



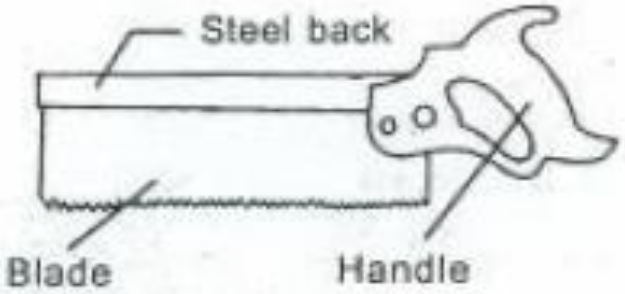
Cutting Tools

a – Cross cut saw

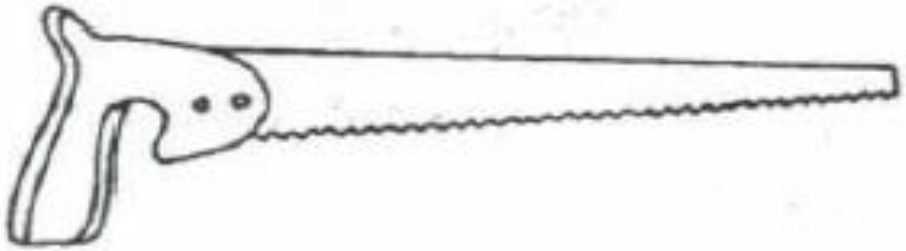
b – Cross cut saw teeth

c – Rip saw teeth

Cross-cut and rip saw



Tenon saw



Compass saw

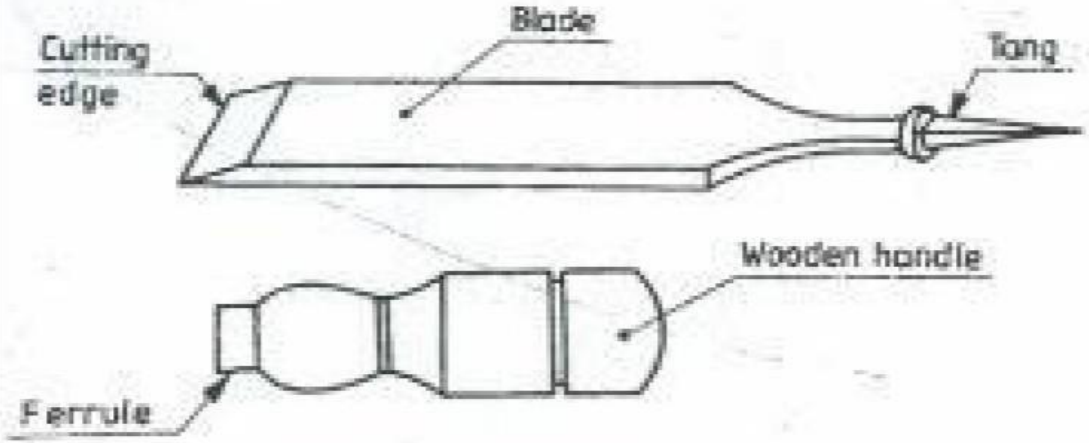


Fig. 1.14 Parts of chisel

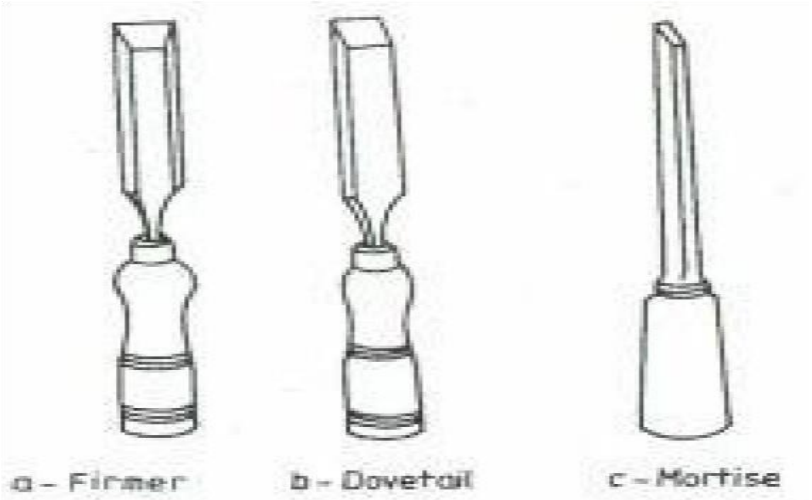


Fig. 1.15 Chisels

Drilling and Boring Tools

Fig. 1.16 Carpenter's brace

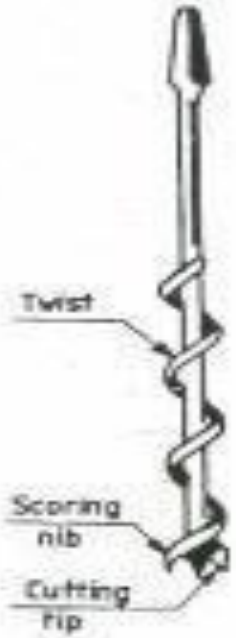


Fig. 1.17 Auger bit

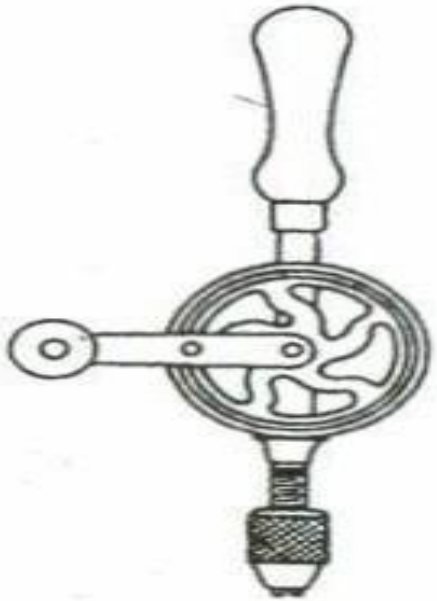


Fig. 1.18 Hand drill

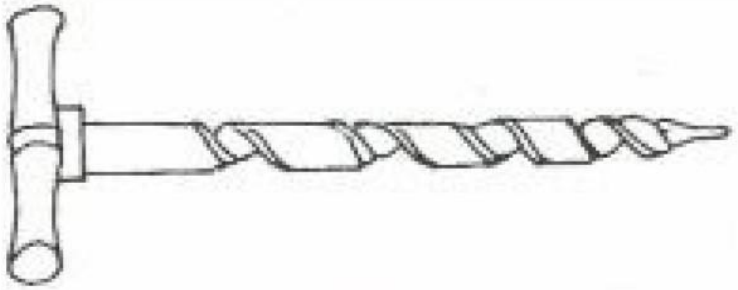
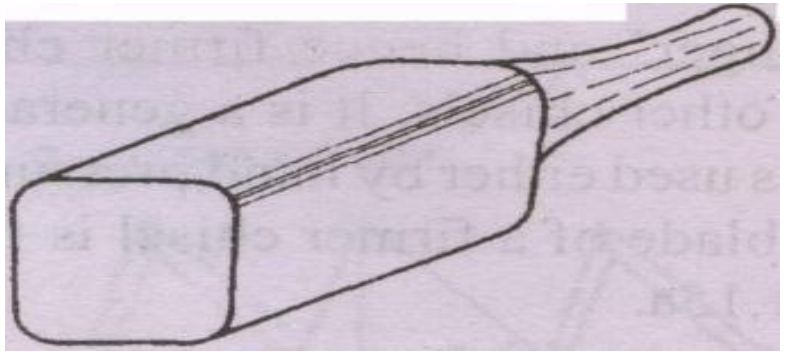


Fig. 1.19 Gimlet

Miscellaneous Tools



Mallet

Pincer

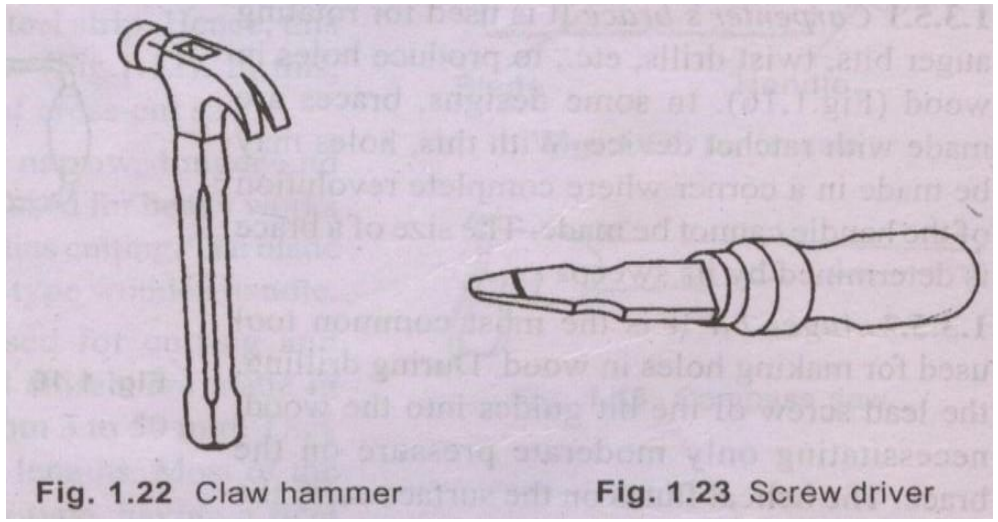


Fig. 1.22 Claw hammer Fig. 1.23 Screw driver

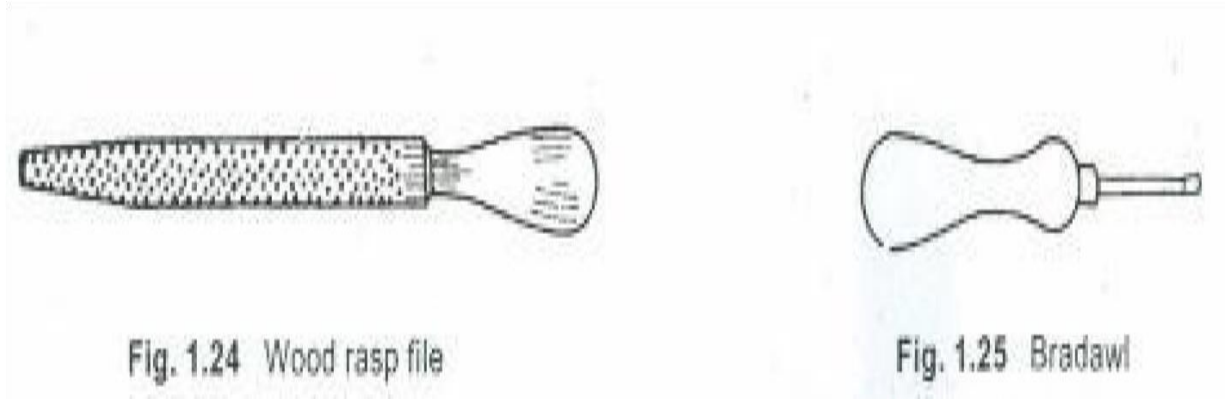
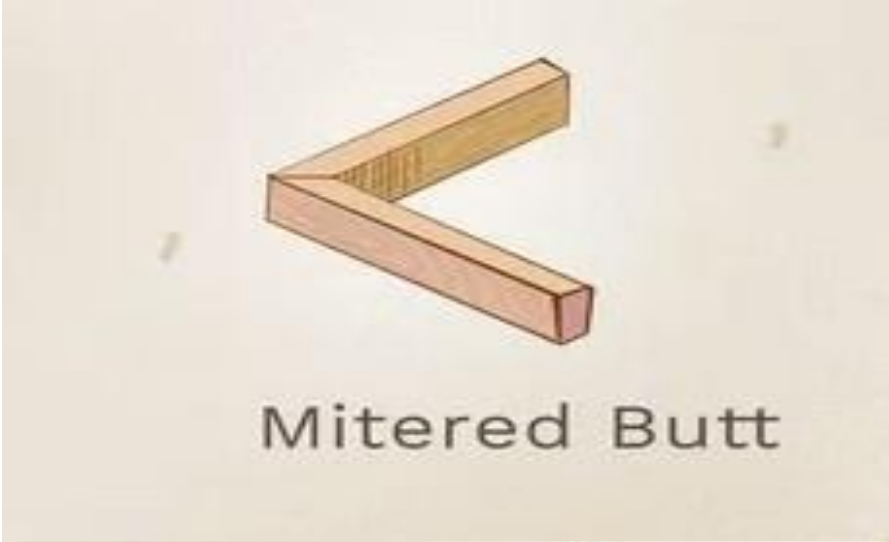
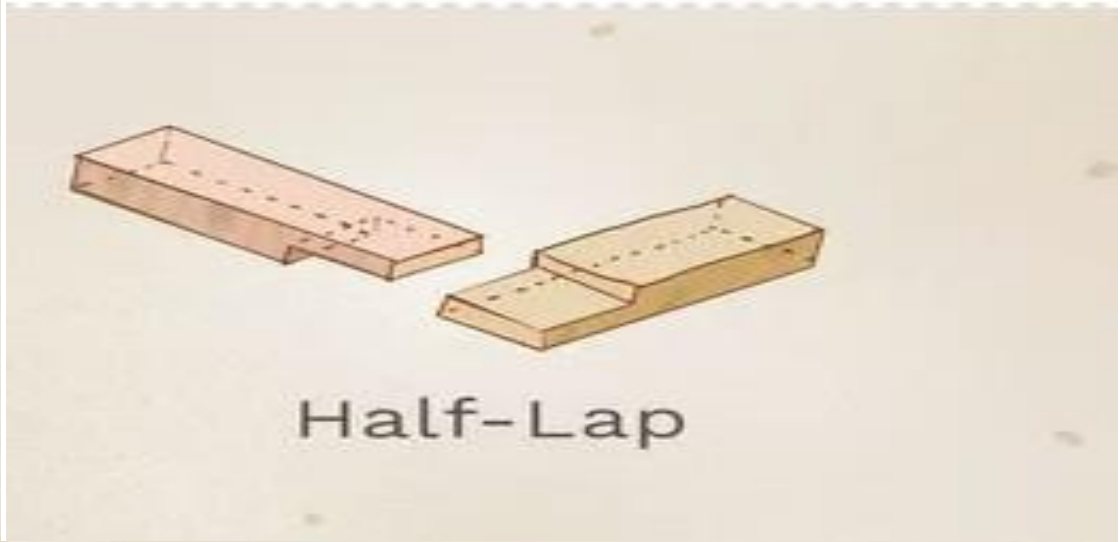
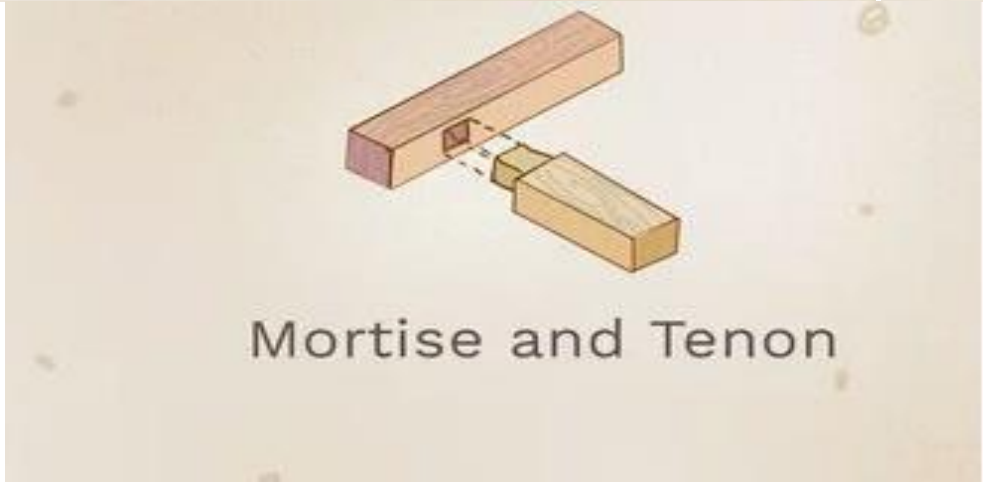


Fig. 1.24 Wood rasp file Fig. 1.25 Bradawl

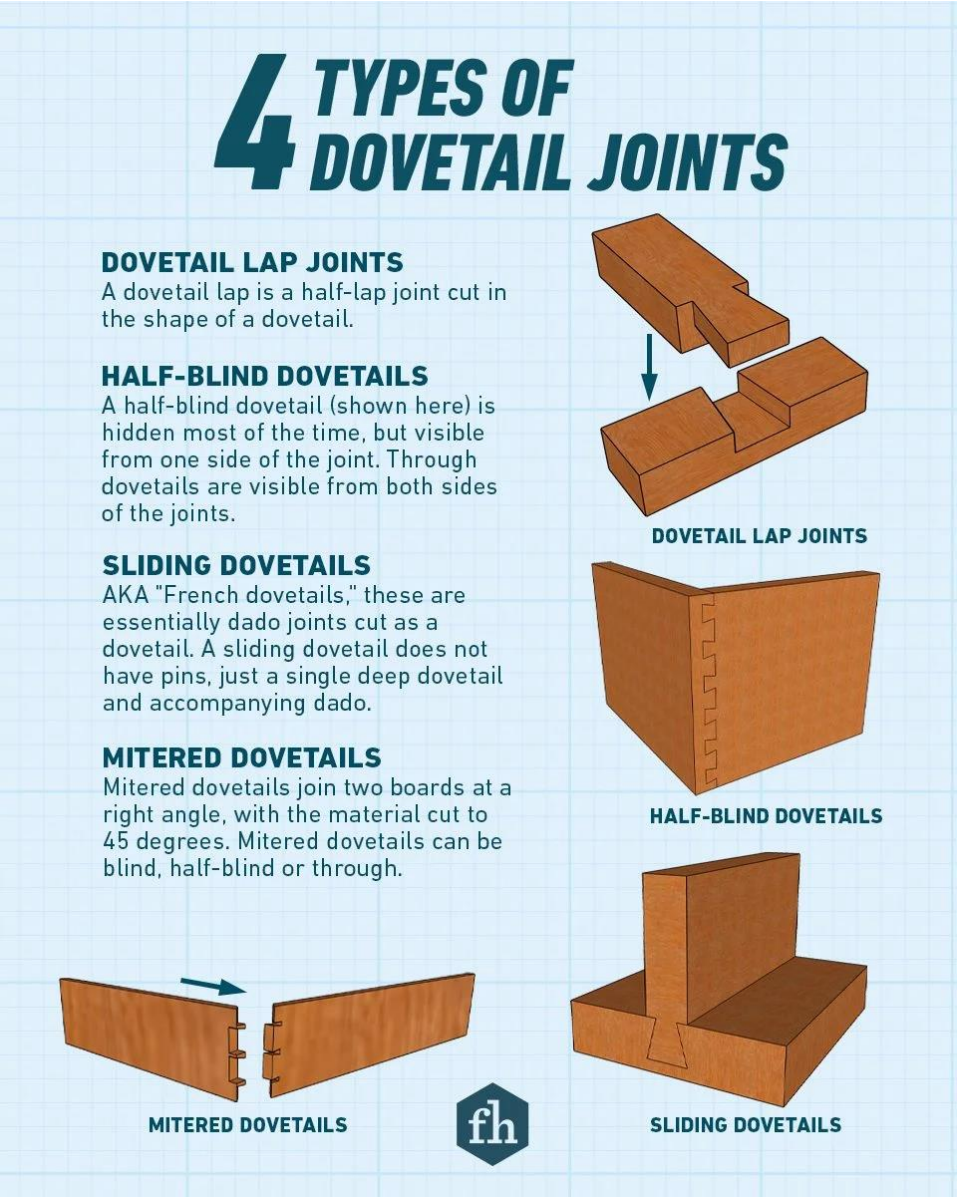
Wood Joints







Pocket Joint


4 TYPES OF DOVETAIL JOINTS
DOVETAIL LAP JOINTS
A dovetail lap is a half-lap joint cut in the shape of a dovetail.
HALF-BLIND DOVETAILS
A half-blind dovetail (shown here) is hidden most of the time, but visible from one side of the joint. Through dovetails are visible from both sides of the joints.
SLIDING DOVETAILS
AKA "French dovetails," these are essentially dado joints cut as a dovetail. A sliding dovetail does not have pins, just a single deep dovetail and accompanying dado.
MITERED DOVETAILS
Mitered dovetails join two boards at a right angle, with the material cut to 45 degrees. Mitered dovetails can be blind, half-blind or through.
DOVETAIL LAP JOINTS
HALF-BLIND DOVETAILS
MITERED DOVETAILS
fh
SLIDING DOVETAILS